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Unilateral environmental policy and offshoring☆

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ABSTRACT

This paper examines how offshoring shapes emissions leakage and global emissions in response to unilateral environmental policy. We use a general equilibrium offshoring model with heterogeneous firms, based on standard modeling assumptions, where firms allocate labor between production tasks and emissions abatement. We find that global emissions respond non-monotonically to a unilateral emissions tax increase: for small cross-country tax differentials, emissions fall; but as the difference widens, leakage exceeds 100%, raising global emissions due to a global technique effect. The cleanest domestic firms start offshoring and incumbent offshoring firms become dirtier under declining effective foreign taxes. We isolate the offshoring margin – one underexplored channel of leakage – and contrast our findings with comparable models of trade in final goods: the mode of globalization matters. Complementing the unilateral reform with a border carbon adjustment (BCA) prevents emissions leakage but may raise inequality between countries.

1. Introduction

Offshoring – the relocation of production abroad – raises fundamental questions about its environmental and distributional consequences. It allows firms to exploit cost differences and avoid stricter environmental regulations – emissions leak, jobs leak, income leaks. This practice may reflect firms' responses to tightening climate policies in countries committed to the Paris

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Agreement (UNFCCC, 2015), as widening disparities in carbon pricing (OECD, 2021a) provide stronger incentives for firms to shift production to lower-regulation regions.[1]

A long-standing question in the field of trade and the environment is to what extent an open economy can unilaterally implement ambitious environmental policies without merely shifting pollution abroad. The existing literature has extensively examined emissions leakage through trade in final goods and energy, emphasizing shifts in comparative advantages and the loss of competitiveness in energy-intensive sectors (e.g., Felder and Rutherford, 1993; Aichele and Felbermayr, 2015). However, this perspective does not account for a key dimension: the mode of globalization matters. Our paper shifts the focus to offshoring, demonstrating how the environmental consequences of unilateral environmental policy depend on the relocation of emissions-intensive tasks across borders. By isolating this particular mechanism, our model provides a partial – but analytically transparent – view of the broader problem of emissions leakage.[2] Building on conventional modeling approaches in trade theory, we examine what we can learn about firm-level pollution decisions, emissions leakage, and the impact of unilateral environmental policy from standard assumptions in a model with offshoring.

We develop a general equilibrium model where heterogeneous firms allocate labor across production tasks and emissions abatement, integrating key elements from Egger et al. (2015) and Copeland and Taylor (1994, 2003). Firms self-select into offshoring, with only the most productive firms able to relocate their emissions-intensive production stage to countries with lower environmental and or labor costs.

This selection mechanism leads to a counterintuitive result: a unilateral increase in an emissions tax can raise global emissions instead of reducing them. While emissions in the reforming country decline, global emissions exhibit a U-shaped pattern – when initial tax differentials are small, global emissions fall, but as the gap widens, emissions leakage exceeds 100%, outweighing domestic reductions. In such cases, the environmental policy reform becomes counterproductive and harmful to the reforming country when transboundary pollution is the concern.

This paradox arises because the cleanest domestic firms offshore, while incumbent offshoring firms increase emissions due to a declining effective emissions tax in the non-reforming country in general equilibrium. The intuition is as follows. A unilateral emissions tax increase raises domestic production costs of the emissions-intensive input, making offshoring more attractive. As a result, the largest, most productive, and least emissions-intensive domestic firms relocate. These are the firms that can afford to avoid both the emissions tax and abatement costs. Consequently, the share and mass of firms that pay the emissions tax – the tax base – shrinks. While these remaining firms reduce emissions by adjusting their production processes, offshoring firms respond differently. Higher labor demand in the host country of offshoring raises the local wage rate, thereby lowering the *effective* emissions tax (i.e., the tax-to-wage ratio) abroad. This reduction makes polluting cheaper in the non-reforming country, leading incumbent offshoring firms to import more emissions-intensively produced inputs. This mechanism is central to understanding how unilateral environmental policy can backfire. It also highlights the importance of the reforming country's initial openness: the larger the share of offshoring firms, the more firms are affected by a declining *effective* emissions tax rather than the intended stricter regulation. These effects accumulate, changing the average emissions intensity through both within- and across-firm adjustments. A global technique effect thus plays a central role in explaining how unilateral environmental policy can unintentionally increase global emissions.[3]

Our finding of a potential increase in global emissions thereby complements (Babiker, 2005). In his oligopoly model, production shifts to non-reforming regions create a pro-competitive effect, leading to leakage rates exceeding 100% when goods are homogeneous and production exhibits increasing returns to scale. In that framework, it is the more-than-proportionate expansion of energy-intensive goods production in non-reforming regions that drives such excessive leakage in response to unilateral environmental policy.[4]

We derive our results in a fundamentally different modeling framework: under monopolistic competition without strategic interactions, heterogeneous firms produce unique varieties of a differentiated good, where emissions arise from a specific input to production. While aggregate intermediates production remains largely stable, the economy-wide scale declines due to reduced *external* returns to scale as the number of available varieties falls. We emphasize how general equilibrium forces and differentiated firm-level adjustments jointly give rise to a decisive technique effect that can produce the unintended environmental outcome.

To assess the robustness of this finding, we extend the model to account for (i) an endogenous adjustment of the host country's emissions tax via an income effect and (ii) energy market spillovers through a globally traded fossil fuel input. Both extensions show

[1] Countries with higher carbon prices tend to be net importers of embodied CO_2, while those with lower prices are net exporters; this outsourcing of pollution to the Global South has intensified over time (cf. OECD, 2021b; Mahlkow and Wanner, 2023; Copeland et al., 2022).

[2] Cherniwchan et al. (2017) argue that "the interaction between trade liberalization, offshoring, and aggregate pollution emissions is complex and currently not well understood". We seek to advance this understanding.

[3] The technique effect is widely recognized as the primary channel through which trade influences aggregate emissions (cf. Copeland et al., 2022; Antweiler et al., 2001) and in the cleanup of manufacturing (cf. Najjar and Cherniwchan, 2021; Shapiro and Walker, 2018; Levinson, 2009). Similarly, Larch and Wanner (2024) identify a prominent role for the technique effect in a quantitative modeling analysis of unilateral withdrawals from the Paris Agreement. We extend this literature by demonstrating that, in an offshoring model, the technique effect becomes the key channel of emissions responses to a unilateral environmental policy reform. In contrast to previous work, we show that when the gap in environmental policy stringency is sufficiently large, the technique effect can lead to an increase in global emissions.

[4] Kortum and Weisbach (2024) show that, under a unilateral production tax, leakage reaches up to 100% in their calibration. The same mechanism implies that leakage can, in principle, exceed 100% when foreign responses are sufficiently elastic: the production tax lowers world energy demand and the world price, inducing a foreign emissions increase that can outweigh the domestic reduction. Lugovskyy et al. (2025) present a scenario of more than 100% leakage focusing on international transportation: the upcoming maritime CO_2 cap can induce a shift towards more carbon-intensive air and land transport, raising overall emissions.

how additional channels can shape emissions leakage, with the key finding of a U-shaped global emissions response holding under plausible conditions.

Our findings highlight a fundamental policy dilemma: (further) diverging levels of environmental stringency – such as widening gaps in carbon prices between countries with ambitious Paris Agreement commitments and others – can severely weaken the effectiveness of unilateral environmental policies in a globalized world. This underscores the need for integrating environmental and trade policy. Countries have adopted various strategies to address the problem of carbon leakage, with *border carbon adjustments* (BCAs) emerging as one of the most widely discussed instruments. By imposing a BCA, a government closes the carbon price wedge between domestic production and production in countries with lower environmental stringency.[5]

To assess the environmental effectiveness and economic costs of a BCA, we extend our model. We show that, in the presence of a BCA, a unilateral rise in the emissions tax no longer gives additional incentives to offshore, effectively closing the underlying channel of emissions leakage and leading to a necessary decline in global emissions. Both purely domestic and offshoring firms reduce their emissions intensity, with offshoring firms producing cleaner due to a higher *effective* emissions tax in the host country.

The presence of a BCA also has implications for the global income distribution. In its absence, a unilateral environmental policy reform reduces income in the reforming country, while increasing income in the non-reforming country, as production relocates. This shift leads to a convergence in income levels and wage rates between the two countries, thereby reducing between-country inequality. By contrast, the presence of a BCA alters these patterns, resulting in income divergence across countries. In the host country, firms adjust their production as if the local emissions tax were increased, even though the host country's environmental policy remains unchanged.[6] As a result, the host country neither attracts routine task production nor gains from additional tax revenue, making the unilateral reform no longer beneficial from its perspective. In the reforming country, however, a BCA leads to a decline in domestic income inequality, provided that BCA-related tax revenues rise and are redistributed as per capita transfers.

We contribute to multiple strands of the literature by addressing a research gap at the intersection of offshoring and unilateral environmental policy.

First, we add to a vibrant literature that integrates emissions generation into general equilibrium trade models with heterogeneous firms, as developed in the seminal work of Melitz (2003). Initiated by Kreickemeier and Richter (2014), the focus is on trade in final goods and the role of firms able to self-select into exporting (cf. Forslid et al., 2018; Shapiro and Walker, 2018; Cherniwchan and Najjar, 2022).[7] In such a framework, Egger et al. (2021b) examine the role of environmental policy comparable to our analysis. In the absence of offshoring, all domestic firms, exporting and non-exporting alike, are directly affected by the unilaterally increased emissions tax. Through international trade, the environmental policy reform spills over to the non-reforming country in general equilibrium. Due to reduced demand from the distorting tax rise, production and, hence, emissions in the non-reforming country decline. Emissions leakage is negative. This is in sharp contrast to our finding of positive emissions leakage in the context of offshoring, possibly exceeding 100%. The mode of globalization matters.[8]

Second, we contribute to the literature on trade in intermediate goods with a focus on offshoring firms. By capturing international spillover effects on labor markets through the relocation of emissions-intensive production, our offshoring model offers a nuanced explanation for patterns of emission leakage connected to unilateral environmental policy. We thereby build on a growing literature on offshoring considering firm heterogeneity (see e.g. Antràs and Helpman, 2004, Antràs et al., 2006, Egger et al., 2021a, or Egger et al., 2024).[9] Our paper contributes to the literature by analyzing a unilateral rise in environmental stringency in the context of offshoring.[10] Among the few theoretical contributions related to this nexus, Schenker et al. (2018) is particularly noteworthy. Expanding on a multi-stage production process in general equilibrium, they link unilaterally increasing environmental stringency to the relocation of production and emissions. By contrast, our framework incorporates firm heterogeneity, allowing us to account for adjustments within and across firms.

Third, we relate to the expanding firm-level literature examining the nexus between offshoring and emissions. Using data on Swedish, French, and Danish firms, respectively, Akerman et al. (2024), Dussaux et al. (2023) and Leisner et al. (2023) document the impact of offshoring on emission intensity and domestic CO_2 emissions. In Akerman et al. (2024), productivity gains from importing intermediates outweigh the emissions reductions from offshoring polluting stages. By contrast, Dussaux et al. (2023) and Leisner et al. (2023) report significant declines in firm-level emission intensities attributable to offshoring. Similar patterns emerge

[5] To tackle carbon leakage and establish a 'level playing field', the EU, for instance, recently introduced the *carbon border adjustment mechanism* (EU CBAM) to complement its emissions trading scheme.

[6] We abstract from policy feedback abroad to isolate the offshoring channel and thereby allow a direct comparison under a unilateral environmental policy reform. A growing literature emphasizes that BCAs may catalyze stricter climate action in trading partners (e.g. Böhringer et al., 2016; Clausing and Wolfram, 2023; Clausing et al., 2024; Beaufils et al., 2024; Clausing et al., 2025).

[7] LaPlue (2019) embeds firm heterogeneity into a trade model of comparative advantages to account for within and across sector responses to policy changes. Complementing the theoretical literature, empirical work provides evidence for an exporter's environmental premium (e.g. Cole et al., 2013; Holladay, 2016; Richter and Schiersch, 2017; Forslid et al., 2018; Banerjee et al., 2021; Rodrigue et al., 2022). Kwon et al. (2023) finds that both exporting and importing firms emit less per unit of output, while Köveker et al. (2025) show that once embodied emissions in sourced intermediates are accounted for, exporters produce more emissions-intensively overall.

[8] This is in the tradition of a literature, which emphasizes differences in labor market outcome. For instance, Hummels et al. (2014) provide evidence on the different wage effects of exporting and offshoring. Egger et al. (2024) rationalize job polarization by offshoring-specific reallocation of labor to low and high productive firms.

[9] Expanding on a broad offshoring survey, Bernard et al. (2025) highlight firm-level differences in relocation incentives. For the role of productivity dispersion in intermediate goods trade, see Bernard et al. (2003).

[10] Several papers assess the impact of environmental policy using comparable models with heterogeneous firms in closed economy setups. See, for instance, Konishi and Tarui (2015), Tombe and Winter (2015), Andersen (2016), Anouliès (2017), Andersen (2018), and Najjar and Cherniwchan (2021).

in Cherniwchan (2017), analyzing US manufacturing firms after NAFTA's implementation.[11] Evidence from Japanese firms links increased carbon content in imports to overall reductions in emissions intensity among firms initiating outsourcing (Cole et al., 2021). For Belgian manufacturing, Michel (2013) attributes a significant share of emissions reductions to offshoring. Offering an analytical perspective, Park et al. (2024) emphasize offshoring's role in mediating the environmental effects of trade liberalization, even absent differences in regulatory stringency. In contrast, our study analyzes a unilateral environmental policy reform under heterogeneous regulatory stringency, explicitly incorporating emissions embodied in offshored intermediates to evaluate leakage. This firm-level evidence on the role of offshoring in shaping emissions outcomes motivates a parallel examination of how environmental policy itself influences firms' offshoring decisions and emissions leakage.

Fourth, we thus add to the relatively small literature examining environmental policies' effects in the context of offshoring. Cole et al. (2017) highlight environmental regulation as a key driver of FDI and potential emissions leakage. Ben-David et al. (2021) show that multinational firms headquartered in stricter regulatory environments generally emit less than those based in less regulated countries, extending country-specific evidence such as Cole et al. (2014) for Japan. Using cross-border M&A data, Saussay and Zugravu-Soilita (2023) also demonstrate sensitivity of corporate restructuring to relative policy stringency. Antonietti et al. (2017) find stricter regulation in the North increases outsourcing to less regulated Southern suppliers, while (Hanna, 2010) links Clean Air Act amendments to higher US outbound FDI without a strong North–South pattern. More recently, Tanaka et al. (2022) provide causal evidence that stricter US regulations induced pollution offshoring in lead–acid battery recycling at the US-Mexican border. By contrast, Dechezleprêtre et al. (2022) find no evidence of carbon leakage from EU ETS-covered firms, suggesting moderate carbon price differentials may not trigger emissions relocation. Our contribution lies in developing a theoretical framework that links stricter unilateral environmental policy to cross-border production shifts and emissions leakage, explicitly modeling offshored emissions.

Lastly, we contribute to the rich and growing literature on the effectiveness of BCAs. Computable general equilibrium (CGE) models generally find that BCAs substantially reduce emissions leakage (Elliott et al., 2010; Fischer and Fox, 2012; Böhringer et al., 2012a,b; Schenker et al., 2018), although some studies report more modest effects (Babiker and Rutherford, 2005; Böhringer et al., 2022). Structural gravity models similarly associate carbon tariffs with significant leakage mitigation, while underscoring potentially large welfare costs (cf. Larch and Wanner, 2017). Farrokhi and Lashkaripour (2025) demonstrate that border taxes offer only limited scope for global emissions reduction – even under optimal design. Building on a framework with heterogeneous abatement costs, Ambec et al. (2024) show that BCAs curb leakage but may distort competition, while (Artuc and Sommer, 2024) highlight that failing to cover emissions embodied in upstream inputs can induce downstream leakage through production outsourcing. Quantitative models featuring heterogeneous firm-level sourcing choices (Coster et al., 2024; Stillger, 2025), and those incorporating both productivity heterogeneity and input–output linkages (Sogalla, 2025), suggest that the effectiveness of the EU CBAM is constrained by firms' flexibility in adjusting supply chains. We add to this literature by analyzing unilateral environmental policy in a setting where heterogeneous firms can offshore emission-intensive stages of production. In our model, a BCA that complements stricter domestic regulation, neutralizes any additional incentives to offshore, thereby fully preventing emissions leakage and potentially reversing the distributional effects of the unilateral policy reform.

The remainder of this paper is structured as follows. Section 2 introduces the model framework and derives the offshoring equilibrium. Section 3 analyzes a unilateral environmental policy reform focusing on the effects on firm selection into offshoring and the factor allocation, on aggregate emissions and emissions leakage, and on income inequality. Section 4 extends our model framework to explore the impacts of a BCA. Section 5 concludes.

2. A general equilibrium model of offshoring with emissions

We consider an economy with a final goods sector and an intermediate goods sector, following Egger et al. (2015). The final good is produced using differentiated varieties of the intermediate product as the sole input and does not generate emissions. The production of intermediates involves two tasks: a non-routine task, which is emissions-free and performed at headquarters, and a routine task, which is emissions-intensive and can be offshored. Firms, consisting of a manager and workers allocated to production and emissions abatement, either produce entirely domestically or offshore part of their production.[12] The offshoring decision weighs fixed costs against the potential profit gains from lower labor and/or emissions costs abroad.

The economy consists of two asymmetric countries: the source country of offshoring, with an exogenous population size N, and the host country of offshoring, with population N^*.[13] In the source country, individuals differ in managerial ability and can choose between occupations (cf. Lucas, 1978). Those who run firms have their managerial ability translated into firm productivity, leading to firm heterogeneity. By contrast, individuals in the host country can only perform the routine task as workers; no final or intermediate goods production takes place.

Income in both countries is solely used to consume the source country's final good, which is freely tradable. Trade is balanced, with the final good shipped in one direction and the output of offshored routine tasks sent in exchange. Final output Y serves as the numéraire, with its price normalized to unity.

[11] Choi et al. (2025) identify reduced US-China trade policy uncertainty as a key driver of emissions reductions via offshoring in US manufacturing, especially in regions with stricter regulation and higher pollution intensity.

[12] We follow Grossman and Rossi-Hansberg (2008, p. 1981), who define offshoring as the international displacement of tasks , either within the firm (in-house) or to external suppliers (foreign outsourcing).

[13] We indicate expressions for the host country with an asterisk.

2.1. The final goods sector

Following Ethier (1982) and Matusz (1996), we define final goods output as a CES aggregate of differentiated intermediate goods $q(v)$:

$$Y = \left[\int_{v\in V} q(v)^{\frac{\sigma-1}{\sigma}}\, dv\right]^{\frac{\sigma}{\sigma-1}}, \tag{1}$$

where V denotes the set of available varieties of the intermediate good, and $\sigma > 1$ is the elasticity of substitution between varieties.

Under assumed perfect competition, profit maximization leads to the demand for each intermediate variety v:

$$q(v) = Y\, p(v)^{-\sigma}. \tag{2}$$

Demand depends positively on joint income of the two countries, which equals total final goods output Y, and negatively on the variety's own price.

2.2. The intermediate goods sector

Firms in the intermediate goods sector operate under monopolistic competition, each producing a unique variety v of the differentiated intermediate good. A firm's productivity is determined by its manager's managerial ability φ, which follows a Pareto distribution with lower bound of one and shape parameter $k > \sigma$.[14] The cumulative density function is given by $G(\varphi) = 1 - \varphi^{-k}$. Each manager chooses worker employment and allocation, offshoring activity, and production volume of her variety v.

We specify the production technology as

$$q(v) = \varphi(v)\left[\frac{x^n(v)}{\eta}\right]^{\eta}\left[\frac{x^r(v)}{1-\eta}\right]^{1-\eta} \qquad \text{with} \quad \eta \in (0.5, 1), \tag{3}$$

where $x^n(v)$ and $x^r(v)$ denote the outputs generated by the execution of the non-routine and routine tasks, respectively, which serve as inputs in the production function.[15]

Task execution follows the same technology regardless of production location. Labor is allocated across tasks, following Acemoglu and Autor (2011) and Egger et al. (2015).[16] Additionally, firms allocate labor to emissions abatement, as in Copeland and Taylor (2003). We describe the technology as follows:[17]

$$x^n = l^n \qquad \text{and} \qquad x^r = \beta\,(e)^{\alpha}\,(l^r)^{1-\alpha}, \quad \text{with} \quad \beta \equiv (1-\alpha)^{-(1-\alpha)}\alpha^{-\alpha}, \quad \alpha \in (0,1), \tag{4}$$

where l^n denotes labor allocated to performing the non-routine task, l^r represents labor allocated to performing the routine task, including abatement, and e corresponds to the emissions generated. As common in the literature (e.g. Copeland and Taylor, 1994; Shapiro and Walker, 2018; Egger et al., 2021b), emissions are treated as an input factor that is imperfectly substitutable with labor.[18] The formulation entails a declining marginal effectiveness of emissions abatement with parameter α representing the cost share of emissions in routine-task production. Thus, Eqs. (3) and (4) describe a nested Cobb–Douglas structure with labor and emissions as inputs.

Each firm takes factor prices as given and determines input demand to minimize costs, subject to technology constraints. A purely domestic firm faces the economy-wide wage rate w and emissions tax $t > 0$. An offshoring firm, by contrast, employs domestic workers at w for the non-routine task and imports the offshored input at price p^{r*}. The routine task is executed abroad using foreign labor and generating emissions abroad, under identical production technology.

Assuming perfect competition, the host country offers its routine-task output at marginal costs: $p^{r*} = (t^*)^{\alpha}(w^*)^{1-\alpha}$, with $t^* \in (0; t]$.[19] International shipments incur iceberg transport costs $\tau \geq 1$. Hence, to use $x^{or}(v)$ units in production, an offshoring firm must purchase $\tau x^{or}(v)$ units.[20]

A firm will offshore only if marginal cost savings justify it. Using the marginal cost expressions for purely domestic and offshoring firms (derived in Appendix A.2), we define the *marginal cost savings factor of offshoring* as:

$$\kappa \equiv \frac{c^d(v)}{c^o(v)} = \left[\frac{1}{\tau}\left(\frac{t}{t^*}\right)^{\alpha}\left(\frac{w}{w^*}\right)^{1-\alpha}\right]^{1-\eta}. \tag{5}$$

[14] A sufficiently large k ensures positive and finite means for relevant model variables.

[15] The lower bound $\eta > 0.5$ implies that the wage rate in the source country is at least as high as in the host country, provided that N^* is not significantly smaller than N (a result derived in equilibrium below). This assumption follows Egger et al. (2015) and is supported by empirical evidence (cf. Baldwin and Dingel, 2022).

[16] By modeling production as a combination of routine and non-routine tasks, we adopt the taxonomy established by Becker et al. (2013). Task differentiation has increasingly gained relevance in offshoring frameworks. Carluccio et al. (2019) present empirical evidence for offshoring-induced changes in skill composition (and thus task assignment) of domestic labor employment.

[17] Since intermediate goods firms may source task outputs externally, we omit the index v on the supply side. However, we attribute both employed labor and generated emissions during task execution to demanding firm v, including embodied labor and emissions from imported routine-task output in offshoring firms.

[18] Details on the emissions-generating process and the abatement technology are provided in Appendix A.1, following Copeland and Taylor (2003).

[19] We restrict the model to the empirically plausible case where the host country's emissions tax does not exceed that of the source country.

[20] In the given framework, iceberg transport costs are equivalent to assuming lower (Hicks-neutral) productivity for routine-task production abroad.

An offshoring equilibrium materializes if and only if $\kappa > 1$, meaning offshoring generates sufficient cost savings. The incentive to offshore is driven by two key factors: *(i.)* an across-country environmental tax differential and *(ii.)* an across-country wage gap. Transport costs, by contrast, reduce the incentive to offshore. While emissions taxes t and t^* and transport costs τ are exogenous, wages w and w^* adjust endogenously to policy changes. We explore this feature in Section 3, focusing on a unilateral environmental policy reform.[21]

Equipped with these insights, and noting that firms in the intermediate goods sector charge a constant markup $\sigma/(\sigma-1) > 1$ over marginal costs, we express firm differences in operating profits as:[22]

$$\frac{\pi^o(\varphi)}{\pi^d(\varphi)} = \kappa^{\sigma-1} \qquad \text{and} \qquad \frac{\pi(\varphi_1)}{\pi(\varphi_2)} = \left(\frac{\varphi_1}{\varphi_2}\right)^{\sigma-1}. \tag{6}$$

Thus, offshoring enables a firm to earn higher operating profits (as further discussed in the context of firm selection below). Within each firm type – purely domestic or offshoring – a more productive firm generates higher operating profits. In both cases, this advantage arises from larger sales volumes at lower prices.[23]

Additionally, we analyze differences in emissions across firms, accounting for embodied emissions of offshoring firms to ensure a consistent comparison of environmental footprints.

We express the impact of a firm's offshoring status on its total emissions and emissions intensity, defined as the ratio of (direct or embedded) emissions to physical output, i.e., $i(\varphi) \equiv e(\varphi)/q(\varphi)$.[24] Accordingly,

$$\frac{e^o(\varphi)}{e^d(\varphi)} = \frac{t}{t^*}\kappa^{\sigma-1} \qquad \text{and} \qquad \frac{i^o(\varphi)}{i^d(\varphi)} = \frac{t}{t^*}\kappa^{-1}. \tag{7}$$

Hence, offshoring is associated with higher emissions,[25] whereas its impact on emissions intensity remains ambiguous.

If emissions taxes are equal across the countries ($t = t^*$), offshoring only occurs if wage differences ($w > w^*$) are sufficiently large, offsetting transport costs τ, to ensure $\kappa > 1$. In this case, the tax–wage ratio, i.e., the *effective* emissions tax, is necessarily higher in the host country, and offshoring firms operate at a lower emissions intensity. This logic extends to cases with $t > t^*$, provided the tax differential is sufficiently small. Accordingly, even in such a pollution haven type scenario, offshoring may still reduce emissions intensity. Conversely, if tax differentials are large and drive offshoring, offshoring firms exhibit higher emissions intensity.

The ambiguity in the ratio of emissions intensities contrasts with models of heterogeneous firms selecting into exporting: controlling for productivity, exporters either produce equally emission-intensively as purely domestic competitors, since they adjust to the same domestic tax–wage-ratio (cf. Egger et al., 2021b), or are characterized by lower emissions intensities due to higher abatement investments (cf. Forslid et al., 2018).

Finally, for a given firm type, more productive firms emit less per unit of output but generate more emissions due to their larger scale:[26]

$$\frac{e(\varphi_1)}{e(\varphi_2)} = \left(\frac{\varphi_1}{\varphi_2}\right)^{\sigma-1} \qquad \text{and} \qquad \frac{i(\varphi_1)}{i(\varphi_2)} = \left(\frac{\varphi_1}{\varphi_2}\right)^{-1}. \tag{8}$$

We summarize these findings in the following lemma:

Lemma 1.

(i) Conditional on productivity, a firm's emissions intensity is lower in the case of offshoring if the source country's emissions tax is equal to or only slightly higher than that in the host country. A firm's emissions intensity is higher in the case of offshoring otherwise.

(ii) Conditional on firm type (purely domestic or offshoring), a more productive firm generates more emissions but produces less emissions-intensively.

Proof. See Appendix A.3 and the analysis in the text. □

[21] In later extensions – see Section 3.5 – we endogenize the emissions tax in the host country via an income effect and allow emissions costs to depend on an endogenous energy price.

[22] Since firm performance differences depend solely on productivity differentials, we suppress the firm index v and use productivity φ to distinguish firms.

[23] These results follow directly from the cost minimization conditions derived in Appendix A.2. In Online Appendix X.1, we further derive different labor employment ratios between purely domestic and offshoring firms.

[24] Throughout the paper, we define firm-level emissions intensity as emissions per unit of physical intermediate output, which directly aligns with the firm's production technology. We acknowledge, however, that many empirical applications report emissions intensity relative to sales. In our model, this distinction matters because more productive firms set lower prices. As a result, emissions per unit of sales are invariant across firm productivity levels and differ only with respect to a firm's offshoring status. A formal derivation of emissions intensity per unit of sales is provided in the Online Appendix X.2. See also (Sogalla et al., 2024) for a discussion. When analyzing emissions tax-induced changes at the aggregate level, we express emissions intensity relative to final good output Y, which corresponds to both physical quantity and value due to our choice of the numéraire. See Section 3 below.

[25] This directly follows from $t \geq t^*$ (by general assumption) and $\kappa > 1$, a precondition for an offshoring equilibrium, and is driven by the higher output under offshoring.

[26] For empirical evidence on the negative relationship between firm productivity and emissions intensity, see, for instance, Kwon et al. (2023), Forslid et al. (2018), and Richter and Schiersch (2017).

These mechanisms already highlight a core trade-off in how firms can respond to a rise in the domestic emissions tax – namely, between internal abatement and the offshoring of the emissions-intensive routine task (cf. Cherniwchan et al., 2017, p. 77). We formalize this trade-off in the following by characterizing the offshoring equilibrium and examining the effects of a unilateral environmental policy reform.

2.3. The offshoring equilibrium

We now turn to the conditions under which offshoring emerges endogenously in equilibrium.

In the source country, individuals can become managers, workers or offshoring consultants, choosing their occupation solely based on income. If an individual becomes a worker or offshoring consultant, her managerial ability remains unexploited. Accordingly, all non-managers are homogeneous and paid the same economy-wide wage w. By contrast, as a manager (and assumed firm owner), an individual's managerial ability translates into firm productivity, and hence determines the individual's profit income.

In addition to wage payments (for workers and offshoring consultants) or profit income (for managers), each individual receives a uniform per capita transfer $b \equiv tE/N$ from redistributed tax revenues from aggregate domestic emissions E.[27]

Let φ^d denote the threshold managerial ability to run a firm. Individuals with $\varphi \geq \varphi^d$ become managers, while those with $\varphi < \varphi^d$ become workers or offshoring consultants. Consistent with empirical evidence on self-selection into offshoring (cf. Paul and Yasar, 2009; Hummels et al., 2014; Cole et al., 2014), the least productive firm, i.e., the firm run by the marginal manager, does not offshore. This leads to the following condition of the individual who is just indifferent between becoming a manager and choosing a wage-remunerated occupation:

$$\pi^d(\varphi^d) + b = w + b, \tag{9}$$

where the per capita transfer does not influence the decision.

Each manager also decides whether to offshore or produce purely domestically. Offshoring induces higher operating profits from lower variable production costs, see Eq. (8), but requires hiring one offshoring consultant as a fixed input requirement. Accordingly, only the most productive firms offshore. The marginal offshoring firm with productivity φ^o is determined by

$$\pi^o(\varphi^o) - \pi^d(\varphi^o) = w, \tag{10}$$

where the additional operating profits from offshoring (left-hand side) must cover the fixed costs of hiring an offshoring consultant (right-hand side).

Jointly, the two cutoff productivity levels, φ^d and φ^o, determine the share of offshoring firms:

$$\chi \equiv \frac{1 - G(\varphi^o)}{1 - G(\varphi^d)} = \left(\frac{\varphi^d}{\varphi^o}\right)^k. \tag{11}$$

This variable plays a central role in our analysis, as all aggregate variables can be expressed as functions of χ. In the next section, we emphasize its role in evaluating the impact of a unilateral environmental policy reform.

An offshoring equilibrium requires at least some firms to offshore, i.e., an interior solution with $\chi \in (0, 1)$. To establish this, we derive two key relations between the share of offshoring firms χ and the marginal cost savings factor κ, alongside constraints on the exogenous emissions tax differential and transport costs that ensure an interior solution (with derivations in Appendix A.4). For such an interior equilibrium to exist, the emissions tax differential must not be too large, and transport costs must be neither too small nor too large.

The first link between κ and χ, the *offshoring indifference condition* $A(\chi)$, stems from the two indifference conditions Eqs. (9) and (10):

$$\kappa = A(\chi) \equiv \left(1 + \chi^{\frac{\sigma-1}{k}}\right)^{\frac{1}{\sigma-1}}. \tag{12}$$

This equation captures a positive relation between κ and χ: A higher marginal cost savings factor induces relatively more firms to offshore.

The second link between κ and χ, the *labor market constraint* $B(\tau, t, t^*, \chi)$, follows from labor market equilibrium conditions. Given population sizes and occupational choices, the resource constraints for the source country and the host country are:

$$N = L + (1 + \chi)M \qquad \text{and} \quad N^* = L^*, \tag{13}$$

where L and L^* denote the mass of workers, M represents the mass of managers, while χM subsumes all offshoring consultants. From these conditions, we derive relative wage ratio w/w^* as a declining function of χ. Substituting this into the definition of κ in Eq. (5), we obtain:

$$\kappa = B(\tau, t, t^*, \chi) \equiv \left[\frac{1}{\tau}\left(\frac{t}{t^*}\right)^{\alpha}\left(\frac{\gamma^l(\chi)}{\gamma^{l*}(\chi)}\frac{N^*}{\lambda(\chi)N}\right)^{1-\alpha}\right]^{1-\eta}, \tag{14}$$

[27] Redistributing emissions tax revenues as a uniform per capita transfer is a regular policy proposal, as seen in the US Carbon Dividends Plan or the 2022-launched Austrian scheme called "Klimabonus".

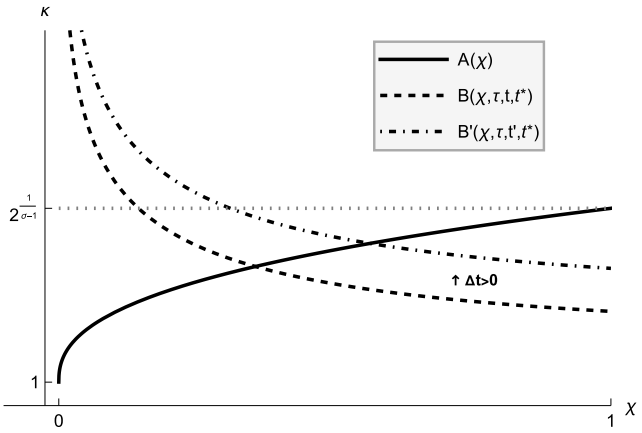


Fig. 1. Determining the offshoring equilibrium
Notes: The solid line shows $A(\chi)$, while the dashed line depicts $B(\tau, t, t^*, \chi)$ in the initial offshoring equilibrium. The dashed–dotted line illustrates the upward shift in $B(\cdot)$ due to the unilateral emissions tax increase, as discussed in Section 3.

where $\gamma^l(\chi)$ and $\gamma^{l*}(\chi)$ denote labor income shares of total factor income in the source country and in the host country, respectively, while $\lambda(\chi) \equiv L/N$ is the share of workers in the source country's population. This equation captures a negative relation between κ and χ: A higher share of offshoring firms (χ) increases labor demand in the host country, while reducing it in the source country. This, in turn, causes a relative rise in the host country's wage rate, making offshoring less attractive and leading to a decline in κ.

Fig. 1 illustrates an interior offshoring equilibrium, where the solid line represents the upward-sloping $A(\chi)$ function, Eq. (12), and the dashed line represents the downward-sloping $B(\tau, t, t^*, \chi)$ function, Eq. (14). The intersection of these curves determines the equilibrium values of κ and χ. The dashed–dotted line will be introduced in the next section.

3. Unilateral environmental policy reform

This section analyzes the effects of a unilateral environmental policy reform in an offshoring equilibrium. Specifically, we examine an increase in the source country's emissions tax, while the host country's tax remains unchanged in our default case, i.e., $dt > dt^* = 0$. This assumption implies that while t changes in real terms, t^* remains fixed in terms of the numéraire Y. This setup provides a clean benchmark to characterize the resulting adjustment mechanisms.

Key to our analysis is understanding how the share of offshoring firms, χ, adjusts in response to a rise in t. Building on this, we examine how occupational choices and aggregate factor allocation change (Section 3.1), providing insights into the broader economic impacts. These results are essential for assessing changes in aggregate emissions in the reforming source country and emissions leakage to the non-reforming host country (Section 3.2). We further decompose these changes to identify the key underlying mechanisms (Section 3.3). In addition, we analyze the distributional effects of the reform (Section 3.4).

Finally, we examine the robustness of our results through two extensions that modify how emissions costs in the host country are determined (Section 3.5): (i) an endogenous adjustment of the host country's emissions tax via an income effect, and (ii) a policy-induced change in energy prices through an energy market channel.[28]

3.1. Firm selection into offshoring and factor allocation

A unilateral increase in the source country's emissions tax directly raises domestic production costs and enhances the cost advantage of offshoring, as reflected in a higher marginal cost savings factor, κ. This shifts the firm distribution towards offshoring, i.e., $d\chi/dt > 0$, and alters labor demand across countries, causing the host country's wage to rise relative to the source country. As a result, the increase in κ – and thus in χ – is partially attenuated, while, in general equilibrium, both increase. Fig. 1 illustrates the new offshoring equilibrium, induced by an upward shift from $B(\chi, \tau, t, t^*)$ to $B'(\chi, \tau, t', t^*)$, the dashed–dotted line.[29]

Not only does the share of offshoring firms increase, but so does the total mass of offshoring firms, i.e., $d(\chi M)/dt > 0$. At the extensive margin, more firms avoid paying the domestic emissions tax by offshoring part of their production. These firms are the most productive among previously purely domestic firms, responding to rising production costs by relocating production abroad.

The implication of this result is remarkable: as deduced from Lemma 1, the cleanest purely domestic firms – those with the lowest emissions intensities per unit of physical output – start to offshore in response to an emissions tax increase. This runs against the common perception that the dirtiest firms are the ones most likely to offshore.[30] In our model with heterogeneous firms, this

[28] A fundamentally different policy intervention – the introduction of a border carbon adjustment – is analyzed separately in Section 4.
[29] Formally, this follows from the direct effect of t on $B(\chi, \tau, t, t^*)$ in Eq. (14), while $A(\chi)$ in Eq. (12) remains unaffected.

outcome arises due to the fixed costs of offshoring: only sufficiently productive firms can afford to offshore, and an emissions tax rise shifts the decision in favor of offshoring.

While purely domestic firms at the upper end of the firm distribution switch into offshoring, the effect at the lower end is ambiguous. To formalize this, we define two threshold levels of χ:[31]

First, for $\chi > \tilde{\chi}$, an increase in the source country's emissions tax reduces the mass of active firms, $dM/dt < 0$. The least able managers stop running a firm. This also implies a decline in the mass of purely domestic firms, $d(1-\chi)M/dt < 0$, as they exit on both ends of the firm distribution, into inactiveness and offshoring. Second, for $\tilde{\chi}^d < \chi < \tilde{\chi}$, the mass of active firms increases, $dM/dt > 0$, despite a decline in purely domestic firms, $d(1-\chi)M/dt < 0$. Low-productivity firms enter the market, but are outnumbered by relatively high-productive firms that start offshoring. Finally, for $\chi < \tilde{\chi}^d$, both the mass of purely domestic firms and the mass of all active firms increase.

This implies that when the initial share of offshoring firms is low ($\chi < \tilde{\chi}$; second and third case), low-productivity, highly emissions-intensive firms are newly established. In these cases, the small group of offshorers cannot absorb all of the labor freed up by the emissions cost shock. As a result, some relatively low-ability individuals newly become managers (cf. Egger et al., 2015), leading to an increase in the mass of emissions-intensive firms.

This result contrasts fundamentally with Melitz-type models of trade in final goods, where a stricter unilateral environmental policy unambiguously leads to the exit of the least productive firms, which are also the most emissions-intensive. In those models, exporters – being relatively less affected due to a smaller demand reduction in the non-reforming country – absorb the freed labor of exiting firms (cf. Egger et al., 2021b). In contrast, in the context of offshoring, newly offshoring firms replace domestic workers performing the routine task with foreign labor.

Regardless of the ambiguity in M, an increase in t unambiguously reduces the total mass of workers in the source country, L. This decline in labor demand is driven by two forces: higher production costs, which lower input demand, and increased offshoring, as offshoring firms employ domestic workers only for non-routine tasks.

Zooming in on the reduction in domestic labor demand, we observe two important tax-induced labor shifts in the source country. First, workers reallocate from the routine to the non-routine task, both in relative and absolute terms.[32] Second, labor moves from purely domestic firms to offshoring firms, even though the latter execute only the non-routine task within the source country. These effects are formally captured by the following labor reallocation responses to a rise in t:

$$\frac{dL^d}{dt} < \frac{dL^r}{dt} < \frac{dL}{dt} < 0 < \frac{dL^n}{dt} < \frac{dL^o}{dt}. \tag{15}$$

These reallocation patterns in response to the emissions tax affect general equilibrium wage rates. In the source country, the decline in total labor demand puts downward pressure on the domestic wage, w. In contrast, the expansion of offshored production raises labor demand in the host country, increasing w^*.

We summarize these findings as follows:

Proposition 1.

A unilateral increase in the source country's emissions tax leads to:

(i) an increase in the share and mass of offshoring firms, driven by the most productive, least emissions-intensive domestic firms.

(ii) a decline in the mass of workers, with labor reallocated towards offshoring firms and shifted from routine tasks to non-routine tasks. The effect on the mass of active firms is ambiguous: it declines when the initial offshoring share is high, but increases otherwise.

(iii) a reduction in the wage rate w in the source country and an increase in the wage rate w^* in the host country, implying a reduced relative wage w/w^*.

Proof. See Appendix B.1. □

3.2. Aggregate emissions changes

Equipped with these insights, we now examine how a unilateral increase in the source country's emissions tax affects emissions. Aggregate emissions in the source country are given by:

$$E = \frac{\gamma^e(\chi)}{t}\frac{\sigma-1}{\sigma}Y, \tag{16}$$

[30] This statement merits qualification along three dimensions. First, our single polluting-sector model isolates the within-sector margin and does not capture cross-sectoral differences in emissions intensity, which may also shape offshoring responses. Second, as discussed above and in Online Appendix X.2, our default measure of emission intensity is based on physical output rather than sales. Purely domestic firms that begin offshoring in response to the emissions tax are equally emissions intensive per unit of sales as those that remain non-offshoring. This implies that, even when emissions are measured relative to sales, offshoring is not selectively undertaken by the dirtiest firms within our framework. Third, the result reflects the standard modeling assumption of one-dimensional Hicks-neutral productivity heterogeneity. Different patterns might arise under alternative forms of firm heterogeneity.

[31] Appendix B.1 derives the two threshold levels, $\tilde{\chi}$ and $\tilde{\chi}^d$, in terms of model parameters. See also (Egger et al., 2015) for a related discussion of an ambiguous effect of a change in trade costs on the mass of firms.

[32] For reasons of tractability, our model does not distinguish between skill levels required for different tasks, resulting in a single economy-wide wage paid to all workers. Outside the model, the finding of a shift of the labor force towards non-routine tasks is consistent with empirical findings on the heterogeneous effects of offshoring across worker types (cf. Hummels et al., 2014).

where $\gamma^e(\chi)$ represents the share of aggregate income, net of operating profits, that accrues to emissions tax payments. Since both γ^e and Y decline with t – in addition to its direct effect – aggregate emissions in the source country fall: $dE/dt < 0$.

This reduction works twofold: directly, the tax discourages emissions-intensive production; indirectly, it raises the share of offshoring firms, as shown in Proposition 1, part (i). The indirect effect operates through two opposing forces: while offshoring mitigates the decline in output Y, it also reduces $\gamma^e(\chi)$ as emissions-intensive activity leaves the country – a pollution-shifting effect.

This outcome contrasts with standard Melitz-type models featuring final goods trade, where such emissions relocation does not occur. In those models, the indirect effect – namely, a tax-induced shift towards international activity in the form of exporting – attenuates the emissions reduction by weakening the increase in the effective emissions tax in general equilibrium (cf. Egger et al., 2021b).

To assess the full environmental impact of a unilateral environmental policy reform, we must also consider emissions in the host country. Despite the decline in overall economic activity Y, aggregate emissions abroad increase: $dE^*/dt > 0$. Firms shift production from the source to the host country to avoid domestic taxation and abatement costs. The rise in emissions is driven by a decline in the host country's effective emissions tax, t^*/w^* – as w^* increases in general equilibrium, see Proposition 1, part (iii). As a result, generating emissions abroad becomes relatively cheaper.

While the existence of positive emissions leakage is not novel, it differs fundamentally from leakage in comparable models of trade in final goods (cf. Egger et al., 2021b). In those models, negative emissions leakage arises as the reforming country's reduced market size lowers production and emissions in the non-reforming country. By contrast, in our offshoring framework, firms actively relocate emissions-intensive production, leading to positive emissions leakage. The mode of globalization matters.

With emissions falling in the source country but rising in the host country, the net effect on global emissions, $E^W \equiv E + E^*$, is *a priori* ambiguous. Remarkably, we find that emissions leakage can exceed 100%, implying that emissions abroad rise by more than they fall at home.

To establish this result, we express global emissions as:

$$E^W = \left(\frac{E}{E^*} + 1\right) E^*, \qquad \text{where} \quad \frac{E}{E^*} = \frac{1 - \chi^{\frac{k-\sigma+1}{k}}}{\chi + \chi^{\frac{k-\sigma+1}{k}}} \frac{t^*}{t}. \tag{17}$$

The emission ratio E/E^* declines with both the share of offshoring χ and the tax ratio t/t^*. Whether global emissions rise or fall then depends on the relative strength of this relocation and the increase in E^*, a relationship that turns out to be non-monotonic in t.

From Eq. (17), the extent of emissions leakage hinges on the initial level of χ. When χ is low, the environmental policy reform reallocates production inputs to relatively few, low emission-intensive offshoring firms, thereby reducing global emissions. In contrast, when χ is high, resources shift towards a broader set of relatively emissions-intensive offshoring firms, potentially increasing E^W.

Since χ endogenously rises with t, there exists a threshold tax level beyond which emissions leakage dominates, causing global emissions to increase. This threshold is positively related to trade costs, τ, which structurally constrain the extent of offshoring.[33]

We summarize our findings as follows:

Proposition 2. *A unilateral increase in the source country's emissions tax leads to:*

(i) a decline in aggregate domestic emissions, amplified by the pollution-shifting effect of offshoring.

(ii) an increase in aggregate emissions in the host country, which is inversely related to the host country's effective emissions tax.

(iii) a net increase in global emissions at sufficiently high tax rates, despite a decline in aggregate economic activity. The more open the reforming country, the lower the threshold tax rate at which emissions leakage surpasses 100%.

Proof. See Appendix B.2. □

3.3. Decomposing emissions changes

Decomposing aggregate emissions is a common strategy in the literature to isolate the partial effects of a policy reform and better understand the underlying mechanisms (cf. Grossman and Krueger, 1995; Antweiler et al., 2001; Levinson, 2009; Cherniwchan et al., 2017; Shapiro and Walker, 2018).

We present two complementary decomposition approaches that differ in their level of aggregation and the adjustment margins they emphasize. We quantify the relative importance of these margins through a numerical exercise.

Our first approach focuses on the change in emissions in the reforming source country, E. This top-down, economy-level decomposition attributes changes in emissions to a scale and a technique effect. Within the technique effect, it highlights a distinctive *task-based composition effect*: the reallocation of labor from the (emissions-intensive) routine task to the (clean) non-routine task in response to the emissions tax increase.

[33] To illustrate the non-monotonic effect of the emissions tax on global emissions, numerical simulations are presented in the Online Appendix Y.1. Two representative scenarios – "low χ" and "high χ" – are compared, where parameters are chosen identically across scenarios except for τ, which is higher in the 'low χ' case. These examples are referred to again throughout the following analyses.

Our second approach examines changes in global emissions, E^W, to unpack the forces behind the net global effect. This bottom-up, firm-level decomposition distinguishes between emissions from purely domestic and offshoring firms, capturing both within-firm and across-firm adjustments. This approach clarifies the channels underlying our central result that emissions leakage can exceed 100%.[34]

3.3.1. Decomposition I: Top-down analysis of source country's emissions

Aggregate emissions in the source country can be expressed as the product of the economy-wide scale, Y, and the average emissions intensity – the technique – of domestic production, $\bar{i}^d_Y$:

$$E = Y \cdot \bar{i}^d_Y = \underbrace{Y}_{\text{scale}} \cdot \underbrace{\frac{\alpha}{1-\alpha}\frac{w}{t} \cdot \frac{L^{dr}}{L} \cdot \frac{L}{Q} \cdot \frac{Q}{Y}}_{\text{technique}}, \tag{18}$$

as derived in Appendix B.3. Here, Q denotes the total physical output of the intermediate sector, while Y reflects total final goods production under increasing external returns to scale. Given our choice of numéraire, Y also measures the value-based scale of the economy.

This decomposition reveals how a change in the emissions tax affects domestic emissions through five distinct channels. The first term captures the scale effect, which operates via Y: a tax-induced contraction in the economy's output directly reduces emissions, holding the technique constant. The remaining four terms jointly constitute the technique effect, each reflecting a specific adjustment margin . The second effect, the *pure technique effect* operates via w/t and reflects a tax-induced shift of routine-task labor from production towards abatement, amplified by the decline in the domestic wage – see Proposition 1, part (iii). The third effect, the *task-based composition effect* operates via L^{dr}/L and captures the decline in the share of domestic labor allocated to the routine task – see Proposition 1, part (ii).[35] This margin is the key addition relative to emissions-augmented Melitz-type trade models without offshoring (cf. Egger et al., 2021b). The fourth effect, the *intermediate-sector productivity effect* operates via L/Q and reflects changes in the inverse of aggregate labor productivity in intermediate production. The fifth effect, the *external economies of scale effect* operates via the inverse of the scale of final goods production, which we define as $\psi \equiv Y/Q$.[36]

To assess the relative importance of the different contributing mechanisms, we conduct a numerical decomposition analysis under two representative scenarios – "low χ" and "high χ". Since the scale effect changes little in both cases, we focus on the four components of the technique effect. Results are shown in Fig. 2.[37]

In both scenarios, the pure technique effect emerges as the dominant driver of reduced emissions intensity. The task-based composition effect also contributes to lower emission intensity, especially under "high χ" , where increased offshoring reallocates routine labor towards the non-routine task in offshoring firms. The intermediate-sector productivity effect plays a smaller role, and its sign depends on the initial degree of offshoring: it slightly raises emissions intensity in the "low χ" scenario but reduces it under "high χ", reflecting differing relative changes in labor and intermediate output. Finally, the external economies of scale effect modestly raises emission intensity in both scenarios, driven by the decline in scale economies resulting from fewer varieties.

3.3.2. Decomposition II: Bottom-up analysis of global emissions

To assess the global implications of the unilateral environmental policy reform and uncover the underlying firm-level mechanisms, we now turn to a bottom-up decomposition of global emissions, E^W.

As in Decomposition I, we use aggregate income Y to capture the overall scale of the economy.[38] We then express global emissions as the product of Y and global technique $\bar{i}_Y$, decomposed as:

$$E^W = Y \cdot \bar{i}_Y = \underbrace{\psi \cdot M \cdot \bar{q}}_{\text{scale}} \cdot \underbrace{\bar{i} \cdot \psi^{-1}}_{\text{technique}}, \tag{19}$$

where $\bar{i}$ is the (output-weighted) average emissions intensity – emissions per unit of physical intermediate output – across all active firms M, and $\bar{q}$ their average output.

Since emissions originate in the production of the emissions-intensive routine task, they can be attributed to intermediate-sector firms – represented by the product $M \cdot \bar{q} \cdot \bar{i}$ in Eq. (19). This allows us to isolate the firm-level drivers of emissions intensity before

[34] In the Online Appendix, we develop two additional decompositions of global emissions. Section Y.5 focuses on the global production of the emissions-intensive routine task, isolating changes in the world composition and in the average emission intensity across both countries. Section Y.6 adapts the firm-level decomposition of Forslid et al. (2018) to our setting, emphasizing incumbent firms' responses as well as the extensive margins of selection into offshoring, entry, and exit.

[35] The share of routine labor employed by purely domestic firms remains constant: $L^{dr}/L^d = [(1-\alpha)(1-\eta)]/[1-\alpha(1-\eta)]$. Thus, the shift towards the non-routine task, $d(L^{dr}/L)/dt < 0$, is entirely driven by offshoring firms employing former routine-task labor for their non-offshorable non-routine task.

[36] The fourth and fifth effects together constitute an *aggregate productivity effect*, expressed by L/Y, the inverse of overall labor productivity. In models with homogeneous firms, external economies of scale can be expressed as an increasing function of the mass of firms, M. By contrast, under firm heterogeneity and endogenous selection into offshoring, no closed-form expression exists for ψ, even assuming a Pareto distribution.

[37] For the full decomposition into both scale and technique effects, see Online Appendix Y3.

[38] This definition reflects both global income and total final goods production, which only occurs in the source country, and is hence our preferred measure of scale in both Decompositions I and II.

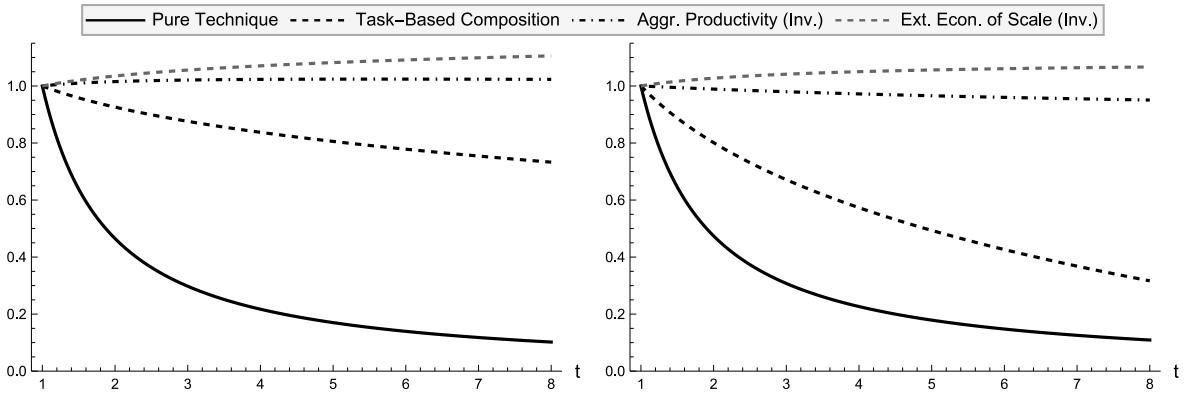


Fig. 2. Decomposition I: The response of the source country's technique $\bar{\imath}_Y^d$.
Notes: The assumed parameter values are $\tau = 2.5$ (for scenario "low χ", left panel), $\tau = 1.2$ (for scenario "high χ", right panel) and $\sigma = 2$, $k = 3$, $\alpha = 0.3$, $\eta = 0.6$, $t^* = 1$, and $N = N^* = 0.5$ (for both scenarios). All illustrated variables are normalized to unity at the level of $t = t^* = 1$.

accounting for scale effects that are external to firms. Accordingly, we make explicit the decomposition $\bar{\imath}_Y = \bar{\imath}\psi^{-1}$, distinguishing between firm-level average and economy-wide emissions intensity.

The firm-level average emissions intensity, $\bar{\imath}$, is a weighted average of the emissions intensities of purely domestic firms, $\bar{\imath}^d$, and offshoring firms, $\bar{\imath}^o$, with weights determined by their relative output shares. For both groups – as derived in Appendix B.3 – average emissions intensity is shaped by within-firm responses to changes in their respective effective tax rates and across-firm selection and reallocation effects, which operate through changes in the average productivity of each group.

For purely domestic firms, the rise in the source country's effective emissions tax, t/w, induces stronger abatement efforts and a reallocation towards the non-routine task.[39] These within-firm adjustments reduce emissions intensity. However, this effect may be counteracted by an *a priori* ambiguous across-firm selection effect, reflected by how the average productivity of domestic firms responds.[40]

By contrast, the average emissions intensity of offshoring firms, $\bar{\imath}^o$, increases unambiguously with the source country's tax rate t. That is, offshoring firms become dirtier on average, despite reallocating towards the non-routine task, as domestic firms do. This occurs because the effective emissions tax in the host country, t^*/w^*, declines endogenously through a general equilibrium rise in the host country's wage; see Proposition 1, part (iii).[41] This effect is reinforced by a fall in average productivity as new, less productive firms begin offshoring.

Moreover, we find that both purely domestic and offshoring firms experience a decline in their average output in response to a higher emissions tax.

We now apply Decomposition II to quantify the global emissions response under the two scenarios introduced above – "low χ" and "high χ". Fig. 3 shows the evolution of E^W, decomposed into scale and technique, as the emissions tax t increases.[42]

In both scenarios, the scale effect reduces emissions throughout, reflecting the contraction in overall economic activity. Yet, despite this continuous downward pressure, aggregate global emissions first fall and then rise – see Proposition 2, part (iii). This pattern implies that the increase in global emissions at higher tax levels must be entirely driven by an increase in average emissions intensity. In other words, the technique effect not only dominates the overall emissions response, but does so in a way that can more than offset the mitigating effect of declining scale.

Essentially, this rise in the technique effect results from a decline in average emissions intensity among purely domestic firms, $\bar{\imath}^d$, and an increase in $\bar{\imath}^o$ among offshoring firms, combined with a reallocation of output towards the latter. The effect is relatively more pronounced in the "high χ" scenario, with the minimum level of global emissions reached at a lower tax differential and a relatively stronger emissions increase thereafter.

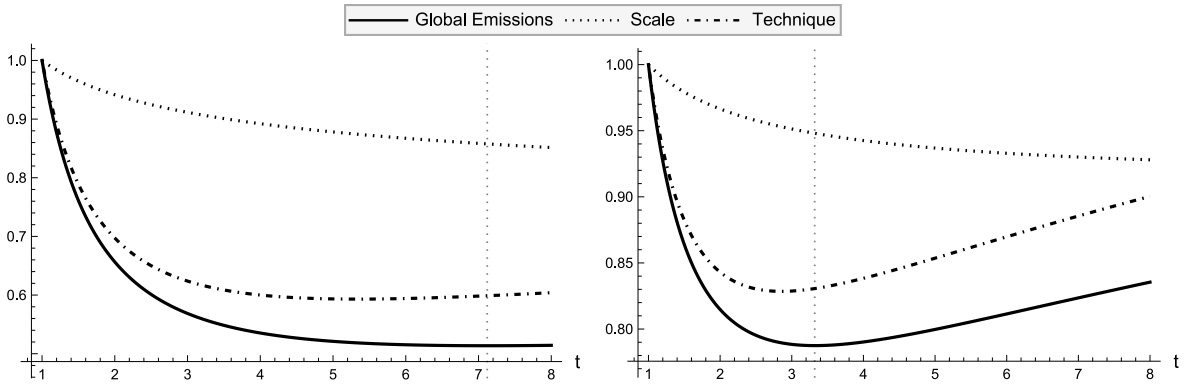
Our finding thereby complements (Babiker, 2005), who reports leakage rates above 100%, driven by a scale effect in an oligopolistic framework with increasing returns to scale. By contrast, our result arises not from output expansion but from a tax-induced rise in the global average emissions intensity – amid a declining scale.

[39] Najjar and Cherniwchan (2021) refer to the within-firm channels of the technique effect as *process effect*. The reallocation of tasks is the firm-level analogue of the *task-based composition effect* in Decomposition I. This adjustment corresponds to the *within-firm reorganization effect* described by Cherniwchan et al. (2017).

[40] As can be inferred from Proposition 1, when the initial share of offshoring firms χ is low, average productivity unambiguously declines due to entry by low-productivity firms and selection of high-productivity firms into offshoring. This may lead to an increase in average emissions intensity $\bar{\imath}^d$, despite within-firm effects. Even in such a case – where a rise in the mass of purely domestic firms $(1-\chi)M$ coincides with an increase in $\bar{\imath}^d$ – aggregate emissions in the source country, E, still decline, as shown in Proposition 2, part (i). This must be due to a dominating decline in average output $\bar{q}^d$.

[41] In Section 3.5, we explore an extension with an endogenous adjustment of the host country's emissions tax via an income effect. This response dampens the decline in the effective emissions tax t^*/w^* and thereby mitigates the rise in $\bar{\imath}^o$, ceteris paribus.

[42] For additional simulation results, see Online Appendix Y4.



(a) Scenario "low χ". (b) Scenario "high χ".

Fig. 3. Decomposition II: The response of global emissions E^W.
Notes: The assumed parameter values are $\tau = 2.5$ (for scenario "low χ", left panel), $\tau = 1.2$ (for scenario "high χ", right panel) and $\sigma = 2$, $k = 3$, $\alpha = 0.3$, $\eta = 0.6$, $t^* = 1$, and $N = N^* = 0.5$ (for both scenarios). All illustrated variables are normalized to unity at the level of $t = t^* = 1$. The dotted vertical lines indicate the emissions tax level that leads to lowest aggregate global emissions in each scenario, respectively.

3.4. Aggregate income and income inequality

Beyond emissions, a unilateral environmental policy reform also affects the distribution of income – both across and within countries.[43] In our stylized setup with limited heterogeneity, these distributional consequences are shaped by the same underlying economic forces that drive emissions leakage. While our focus remains on environmental outcomes, we briefly explore these income effects before returning to the analysis of emissions changes in extended versions of the model in Section 3.5.

World aggregate income declines due to the tax-induced distortion in production. However, this decline is unevenly distributed: as more firms offshore, a larger share of income accrues to the host country. Specifically, income in the host country I^* – comprising worker income and emissions tax revenues – rises in absolute terms. By contrast, aggregate income in the source country I – comprising worker income, emissions tax revenues, and operating profits – declines more than world income overall. Hence, the policy reform induces a shift in income towards the non-reforming country, mirroring the relocation of emissions. This leads to a convergence in aggregate – and, given fixed population sizes, also average – income across countries.

To measure between-country inequality among wage-earners, we define $\Xi \equiv (w + b)/(w^* + b^*)$, where $w + b$ corresponds to the minimum income level in the source country (by the occupational choice condition), while $w^* + b^*$ equals uniform income in the host country. This measure declines in t, i.e., $d\Xi/dt < 0$. A unilateral increase in the emissions tax harms domestic workers relative to those in the non-reforming country, as production shifts abroad and domestic labor demand falls. This complements the earlier result on wage convergence, i.e., a decline in the relative wage w/w^* in response to the increase in t, given that $w > w^*$ – see Proposition 1, part (iii).

Within the source country, individuals earn either the economy-wide wage or firm-profits, with all receiving the transfer b. We define inter-group inequality as $\Theta \equiv (\bar{\pi} - \chi w + b)/(w + b)$, the ratio of average post-transfer managerial income (net of offshoring consultant wage payments) to post-transfer income of workers and offshoring consultants. This measure rises with the emissions tax, i.e., $d\Theta/dt > 0$. Two mechanisms drive this result: First, emissions tax revenue – and thus the transfer b – declines, disproportionately affecting workers. Second, offshoring enables managers to mitigate the tax cost shock, while workers face lower wages amid shrinking domestic labor demand.

We can summarize these findings as follows:

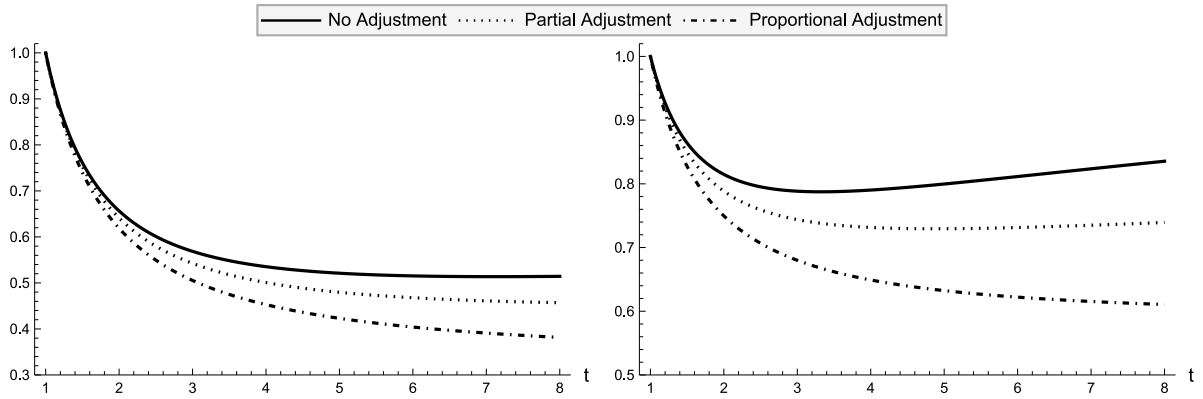
Proposition 3. *A unilateral increase in the source country's emissions tax leads to:*

(i) a convergence in aggregate income between the source and host country, and a decline in between-country inequality accounting for country-specific transfers.

(ii) a rise in inter-group inequality within the reforming source country.

Proof. See Appendix B.4. □

[43] Distributional effects of environmental policy have long been studied and were a prominent topic in the traditional trade and environment literature (cf. Copeland and Taylor, 2003), particularly in relation to changes in relative factor returns. By contrast, distributional effects within new trade theory frameworks are less obvious, as these typically lack meaningful income heterogeneity. Building on an occupational choice framework that generates firm heterogeneity, Egger and Kreickemeier (2012) and Egger et al. (2015) analyze the distributional effects of trade liberalization, while Kohl (2020) and Kohl and Richter (2023) study those of welfare state reforms. We contribute by exploring the impacts of a unilateral environmental policy in the presence and absence of a BCA, as discussed here and in Section 4.



(a) Scenario "low χ". (b) Scenario "high χ".

Fig. 4. Aggregate global emissions with varying elasticities δ.
Notes: The assumed parameter values are $\tau = 2.5$ (for scenario "low χ", left panel), $\tau = 1.2$ (for scenario "high χ", right panel) and $\sigma = 2$, $k = 3$, $\alpha = 0.3$, $\eta = 0.6$, and $N = N^* = 0.5$ (for both scenarios). Values for δ are $\delta = 0$ (no adjustment), $\delta = 1/3$ (partial adjustment) and $\delta = 1$ (proportional adjustment).

3.5. Spillovers and policy responses in the host country

In our baseline model, the unilateral environmental policy reform affects the host country only indirectly: as more firms offshore, the resulting increase in local labor demand raises the wage in the host country, generating a spillover effect of the tax reform.

To check the robustness of our main result – the U-shaped response of global emissions – we consider two extensions that complement this channel. First, we allow the host country's emissions tax to adjust endogenously in response to rising income, which alters the effective emissions tax beyond wage adjustments alone. Second, we introduce an energy market through which global price effects create additional spillovers to the host country. These extensions influence all equilibrium outcomes, including income distribution, but for clarity of exposition we focus on their effects on emissions. Derivations and detailed numerical results for both extensions are provided in the Online Appendix Z.

3.5.1. Income-linked adjustment of host country emissions tax

In this extension, we generalize the baseline model by allowing the host country's emissions tax, t^*, to adjust endogenously to changes in its aggregate income, I^*. This follows the well-established idea in the trade and environment literature that demand for environmental quality rises with income (Copeland and Taylor, 1994, 2003; Antweiler et al., 2001). Higher income – for instance from the attraction of offshored production from the reforming country – induces a welfare-maximizing government to raise environmental policy stringency in response to stronger preferences for a clean environment.

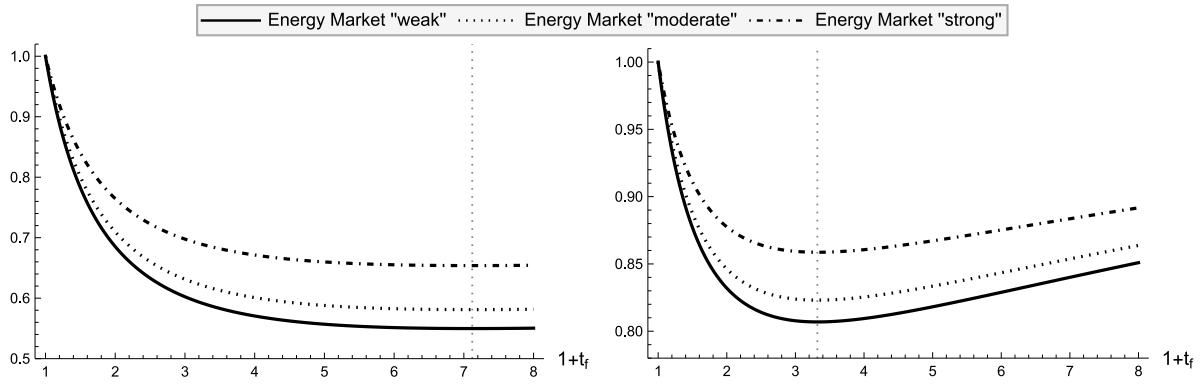
In our framework, host-country income is proportional to its wage, w^*, as labor use is fixed and production of the offshored routine task follows a simple Cobb–Douglas structure. This implies that if t^* adjusts with I^*, it also adjusts with w^*, and thus the tax–wage ratio t^*/w^* varies according to the degree of adjustment. We capture this through an elasticity parameter δ: $\delta = 0$ replicates the baseline model, where the tax is fixed; $\delta = 1$ implies full proportional adjustment, keeping t^*/w^* constant. Intermediate values of δ yield partial adjustments.

An endogenous tax response ($\delta > 0$) moderates offshoring incentives, dampens cross-country general equilibrium effects, and reduces emissions leakage. As a result, the global emissions reduction is more pronounced. For $\delta \in [0, 1)$, a U-shaped pattern for global emissions emerges, with its minimum shifting rightward as the elasticity of adjustment, δ, increases.

Fig. 4 shows how global emissions respond to the unilateral policy reform under varying degrees of host-country tax adjustment, for both "low χ" and "high χ". The three curves correspond to different values of the elasticity parameter δ. The solid line ($\delta = 0$) represents the baseline with a fixed emissions tax t^*. The dashed line illustrates partial adjustment, resulting in a stronger global emissions reduction and a rightward shift in the emissions minimum. This pattern holds for all intermediate values $0 < \delta < 1$, as rising t^* tempers host-country emissions E^* and enhances the decline in source-country emissions E. The dotted line shows the case of proportional adjustment ($\delta = 1$), where global emissions decline monotonically, and the U-shaped response disappears. This reflects that host-country emissions depend solely on the ratio t^*/w^* in our framework, see Proposition 2, part (ii). In sum, the elasticity δ critically shapes the emissions response. Yet as long as $\delta < 1$, the U-shaped pattern of global emissions remains robust.

3.5.2. Energy market spillovers

In this extension, we introduce an international energy market into our framework. While the baseline model abstracts from energy market spillovers, any unilateral environmental reform inevitably affects global energy demand. To capture this and test the robustness of our key findings, we extend the model by linking emissions to the use of a freely tradable fossil fuel in routine



(a) Scenario "low χ". (b) Scenario "high χ".

Fig. 5. Global emissions with varying supply elasticities ν.
Notes: The assumed parameter values are $\tau = 2.5$ (for scenario "low χ", left panel), $\tau = 1.2$ (for scenario "high χ", right panel) and $\sigma = 2$, $k = 3$, $\alpha = 0.3$, $\eta = 0.6$, $t_f^* = 0$, and $N = N^* = 0.5$ (for both scenarios). Values for supply elasticity ν vary from $\nu = 2$ (Energy Market "strong") to $\nu = 5$ (Energy Market "moderate") to $\nu = 10$ (Energy Market "weak"). Setting supply shifter at $\zeta = 0.0078$ ("low χ") and $\zeta = 0.0091$ ("high χ") ensures $r_f^w = 1$ at the baseline. The dotted vertical lines indicate the emissions tax level that leads to lowest aggregate global emissions in each scenario, respectively.

task production. This makes the relevant emissions cost a combination of the country-specific, exogenous emissions tax and an endogenous, uniform energy price determined on the world market.

Specifically, we adjust the routine task production technology of Eq. (4) by modeling emissions arising from fossil fuel input r, so that routine tasks are produced according to $x^r = \beta r^\alpha (l^r)^{1-\alpha}$. Following Larch and Wanner (2024), we assume a constant elasticity fossil fuel supply function, $R^W = \zeta (r_f^w)^\nu$, where ζ is a supply shifter, r_f^w the world energy price, and ν the supply elasticity.[44] The magnitude of ν determines the strength of the energy-market channel: when ν is small, supply is inelastic and r_f^w reacts strongly to demand shifts. A large ν approximates the baseline model without an endogenous energy price component. Firms in the source country face energy costs of $p_f = (1 + t_f) r_f^w$, with t_f denoting the ad valorem emissions tax and an analogous expression applying to the host country.

We revisit the benchmark policy experiment in the extended framework by raising t_f and varying the fossil fuel supply elasticity.[45] As shown in Fig. 5, incorporating energy market dynamics dampens the reduction in global emissions achieved by the unilateral emissions tax increase. Up to the point where global emissions are minimized, the rise in t_f is partially offset by a decline in the world market price of fossil fuels, r_f^w, which moderates the increase in the total energy price faced by domestic producers. This mitigating effect is strongest when the energy supply elasticity ν is low, i.e., when energy market feedback is pronounced. At the same time, the fuel price for producers in the host country declines, resulting in an additional emissions leakage channel. In sum, endogenous energy prices weaken the unilateral reform's impact, but global emissions still exhibit a U-shaped response.

4. Border carbon adjustment as a complementary policy

We now extend our model to assess the role of a BCA in mitigating emissions leakage.[46] A BCA closes the gap between domestic and foreign emissions prices by imposing a tariff on the embodied emissions in imports, with the resulting tax revenues accruing to the implementing country.

In our framework, we model the BCA by redefining the emissions cost for offshored production as $\tilde{t}^* \equiv t^* + \hat{t}$, where $\hat{t} \equiv t - t^* \geq 0$ captures the initial tax differential. Tax revenues of $\hat{t} E^*$ are collected by the reforming source country that implements the policy.[47]

The analysis proceeds in two steps. Section 4.1 analyzes the implementation of a BCA in an initial offshoring equilibrium with a fixed tax differential $t > t^*$ – i.e., "leveling the playing field". Section 4.2 then investigates a unilateral increase in t – as in the baseline analysis of Section 3 – now in the presence of a BCA, allowing us to highlight its role in supporting an ambitious environmental policy reform.

[44] Income from fossil fuel supply does not affect production decisions and resulting emissions, as all income in both countries is spent on the final good.

[45] To make results comparable to the baseline model, we normalize ζ so that $r^W = 1$ in the initial equilibrium for each ν, and vary t_f while keeping $t_f^* = 0$.

[46] While our model addresses emissions more broadly, we use the term "border carbon adjustment" (BCA) following common usage in climate policy. The analysis naturally extends to carbon pricing but can also be interpreted more generally as a border adjustment for any emissions-based regulation of sensibly transboundary pollutants.

[47] In our model, a BCA reflects complete carbon pricing of foreign emissions, as the host country performs only the emissions-intensive routine task for export. We further abstract from resource shuffling across markets and assume full information about the carbon content of trade. This informational requirement poses major challenges for real-world implementations such as the EU CBAM, with alternatives to BCAs discussed in the literature (see, e.g., Campolmi et al., 2025). While stylized, our framework is designed to isolate the impacts of a BCA in the context of offshoring.

4.1. Implementation of a BCA

After the implementation of a BCA, purely domestic and offshoring firms face the same emissions cost, i.e., $\tilde{t}^* = t$. Firms adjust to this change by re-optimizing their input mix, which alters the marginal costs savings factor of offshoring, simplifying the expression in Eq. (5) to $\kappa_{BCA} = [(1/\tau)(w/w^*)^{1-\alpha}]^{1-\eta}$. Intuitively, the emissions tax advantage of offshoring disappears; firms now offshore solely to exploit cross-country differences in labor costs.

This change has two main implications. First, conditional on productivity, offshoring firms are unambiguously less emissions-intensive than purely domestic firms, since both face the same emissions tax rate, but offshoring firms face a higher effective emissions tax abroad due to lower foreign wages – see Lemma 1, part (i). Second, the least productive offshoring firms re-shore. The new equilibrium is characterized by a smaller offshoring cost advantage κ and a lower share of offshoring firms χ.[48]

Production and emissions shift from the host to the source country. As a result, aggregate emissions increase in the source country and decline in the host country. Overall, the implementation of a BCA lowers global emissions. The larger the initial tax gap $\hat{t}$, the greater the global reduction in emissions. This outcome stands in stark contrast to a unilateral emissions tax rise, which – as shown in Section 3 – can lead to higher global emissions. In this sense, implementing a BCA can be a more effective climate policy instrument than unilaterally increasing environmental stringency.

Alongside the fall in global emissions, global income declines. The distributional effects of the BCA are thereby critically determined by the allocation of the BCA-related tax revenues, which accrue to the source country. The BCA increases between-country inequality: the wage gap between the source country and the host country widens, between-country inequality Ξ increases, and the aggregate income ratio I/I^* rises. By contrast, inter-group inequality in the source country, Θ, declines. This is driven by reduced profit opportunities from offshoring for high-ability managers and by the per capita transfer of BCA revenues, which disproportionately benefits low-income wage earners.

We can summarize our findings as follows:

Proposition 4. *The introduction of a BCA by the source country eliminates the cost advantage of offshoring stemming from less stringent environmental policy in the host country. In the new equilibrium,*

(i) the share and mass of offshoring firms decline.

(ii) aggregate emissions rise in the source country but fall in the host country and globally. The emissions intensity of offshoring firms is lower than that of purely domestic firms, conditional on productivity.

(iii) income inequality increases across countries but declines within the source country.

Proof. See Appendix C.1 and the analysis in the text. □

4.2. Unilateral environmental policy reform under BCA

We now examine the effects of a unilateral increase in the emissions tax t in the presence of a BCA. In this case, the tax increase in the source country is mirrored by an identical increase in the emissions cost $\tilde{t}^*$ applied to offshored production in the host country.[49] The absence of an emissions cost differential – in contrast to the baseline analysis in Section 3 – fundamentally alters several earlier results.

First, the tax increase does not affect the incentive to offshore, captured by κ. As a result, both the share of offshoring firms χ and the relative wage w/w^* remain unchanged, and Proposition 1, part (i), does not apply.

Second, the factor allocation within the source country is not affected. Individuals do not revise their occupational choice, and firm composition remains unchanged. Proposition 1, parts (ii) and (iii) therefore do not apply either.

Third, aggregate emissions in the source country still decline, consistent with Proposition 2, part (i), but for different underlying reasons. Unlike in the baseline case, the emissions reduction is not driven by firms shifting production abroad but arises solely from within-firm labor reallocation from production to abatement. The average emissions intensity of purely domestic firms unambiguously declines.

Fourth, in the host country, in the absence of an offshore-induced surge in labor demand, the local wage rate w^* declines in response to the costs shock. This general equilibrium effect amplifies the rise in the effective emissions cost, $\tilde{t}^*/w^*$, thereby reinforcing abatement efforts. As a result, emissions in the host country fall due to both scale and technique effects. Emissions leakage turns negative, and global emissions unambiguously decline – in contrast to Proposition 2, parts (ii) and (iii), where (positive) leakage could exceed 100%.

Fifth, we turn to the impact of a unilateral environmental policy reform on income and income inequality in the presence of a BCA. Global income declines, and income levels between the two countries diverge due to the redistribution of emissions tax revenues from the host to the source country.[50]

[48] For a graphical illustration of the shifts due to the implementation of a BCA, see Appendix Fig. C.1.

[49] Specifically, we analyze $dt = d\hat{t} = d\tilde{t}^* > dt^* = 0$, starting from an initial offshoring equilibrium with $\tilde{t}^* = t \geq t^*$. This case is structurally different to a coordinated tax increase in both countries. For distributional outcomes, it matters that the host country does not retain the full revenues from emissions pricing.

[50] The income divergence mirrors the result from the implementation of a BCA – see Proposition 4, part (iii) – but does not arise from changes in offshoring patterns, which remain fixed.

The response of both inequality measures introduced in Section 3 – the source country's inter-group inequality Θ and between-country inequality Ξ – to a rise in the emissions tax t crucially depends on the transfer of BCA emissions tax revenue. Without this international redistribution, both measures would remain unchanged because the selection into offshoring is unaffected. With the BCA transfer, as long as it increases, between-country inequality Ξ rises, while inter-group inequality Θ decreases; their response to t is a priori ambiguous if the transfer declines.[51] Accordingly, we observe crucial differences from the unilateral environmental policy reform without a BCA, reversing Proposition 3 at least in the case of a rising BCA transfer. Although it reduces domestic inequality and is environmentally effective, a BCA could have adverse economic side effects for developing (host) countries.

We can summarize these results as follows:

Proposition 5. *In the presence of a BCA, a unilateral increase in the source country's emissions tax raises the effective cost of offshored production by the same amount. As a result:*

(i) The share of offshoring firms, the factor allocation, and firm composition remain unchanged.

(ii) Emissions fall in both countries as emissions intensities of all firms decline due to within-firm reallocation of labor towards abatement.

(iii) Aggregate income in the host country falls, leading to across-country income divergence. The effect on income inequality depends on the BCA-related transfer: between-country inequality rises if the transfer increases, while inter-group inequality within the source country declines. If the BCA transfer declines, the response of both inequality measures is a priori ambiguous.

Proof. See Appendix C.2 and the analysis in the text. □

5. Conclusions

This paper investigates the effects of unilateral environmental policy in the presence of offshoring. We use a general equilibrium model with heterogeneous firms to analyze how firm selection into offshoring shapes emissions leakage and the broader environmental and economic consequences of a unilateral environmental policy reform.

Our findings reveal that unilateral environmental policies can have unintended global consequences. While domestic emissions decline, global emissions exhibit a non-monotonic response: when the tax differential between countries is small, global emissions fall; when the tax differential becomes large, by contrast, emissions leakage can exceed 100%, causing a net increase in global emissions. These results arise from standard modeling assumptions about firm heterogeneity, offshoring, and emissions abatement.

A decomposition analysis shows that this result is not driven by an economy-wide scale effect – which is negative throughout – but by a global technique effect, driven by changes in abatement efforts, task-based composition, and firm selection. Offshoring by the most productive, least emissions-intensive firms, together with the weakening of the host country's effective emissions tax, is central to this outcome. In a more open reforming country with many offshoring firms, even small tax differences can trigger large leakage rates, raising global emissions.

These findings contrast with typical Melitz-type trade models of export selection and emphasize that not just the level of globalization, but its mode – offshoring versus final goods trade – matters for the impact of unilateral environmental policy. While providing a partial view on emissions leakage, they illustrate the limits of unilateral environmental policies and the potential role of border carbon adjustments (BCAs) as corrective measures.

In the presence of a BCA, a unilateral emissions tax increase no longer gives an additional incentive to offshore. Instead, both purely domestic and offshoring firms adjust to the higher effective emissions taxes they face, leading to negative emissions leakage and a decline in global emissions. Moreover, combining a stricter unilateral environmental policy with a BCA can reduce domestic inequality but may widen international income disparities by shifting tax revenue from the host country to the reforming country.

Without complementary measures – such as BCAs or globally harmonized carbon pricing – unilateral climate policies risk becoming not only ineffective, but even environmentally harmful. An important direction for future research is to integrate export and offshoring decisions within a unified framework and quantify their relative contributions to global emissions and leakage.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. General equilibrium model of offshoring with emissions

A.1. Emissions abatement

We follow Copeland and Taylor (2003) in specifying the emissions-generating process and the abatement technology by two assumptions. First, we assume

[51] See Online Appendix Z.3 for additional numerical results. As we remain in a range where the BCA transfer increases, the illustrations for both inequality measures show opposing effects between the cases with and without a BCA.

$$x^r = l^r \xi, \tag{A.1}$$

where $\xi \in (0;1)$ is the (endogenous) share of l^r that is employed in the production process, while the share $1-\xi$ is devoted to emissions abatement. Second, we assume that emissions are generated as

$$e = \left(\frac{\xi}{\beta}\right)^{\frac{1}{\alpha}} l^r. \tag{A.2}$$

Solving Eq. (A.2) for ξ and inserting into Eq. (A.1) yields Eq. (4) in the main text.

A.2. Derivation of marginal costs

A.2.1. Purely domestic firms

A firm producing variety v incurs costs of $wl^{dn}(v) + wl^{dr}(v) + te^d(v)$, where the superscript d denotes a purely domestic firm. Minimizing costs subject to the production technology in Eqs. (3) and (4) yields the optimal input demands:

$$l^{dn}(v) = \frac{\eta}{\varphi(v)} \left(\frac{t}{w}\right)^{(1-\eta)\alpha} q^d(v) \tag{A.3}$$

$$l^{dr}(v) = \frac{(1-\eta)(1-\alpha)}{\varphi(v)} \left(\frac{t}{w}\right)^{(1-\eta)\alpha} q^d(v) \tag{A.4}$$

$$e^d(v) = \frac{(1-\eta)\alpha}{\varphi(v)} \left(\frac{w}{t}\right)^{1-\alpha} \left(\frac{w}{t^\alpha w^{1-\alpha}}\right)^{\eta} q^d(v). \tag{A.5}$$

The corresponding marginal cost function then follows as:

$$c^d(v) = \frac{\left(t^\alpha w^{1-\alpha}\right)^{1-\eta} w^\eta}{\varphi(v)}. \tag{A.6}$$

This expression reveals how marginal costs increase in the wage rate w and emissions tax t, and decrease in firm productivity $\varphi(v)$.

Finally, combining Eqs. (A.2), (A.4) and (A.5), allows us to derive the share $\xi(v)$ of routine task labor devoted to production (rather than abatement):

$$\xi(v) = \beta \left[\frac{e^d(v)}{l^{dr}(v)}\right]^{\alpha} = \frac{1}{1-\alpha} \left(\frac{w}{t}\right)^{\alpha}. \tag{A.7}$$

The share of routine task labor used for production thus increases in the (endogenous) wage–tax ratio and the cost share parameter α of emissions in routine-task production.

A.2.2. Offshoring firms

An offshoring firm producing variety v incurs costs of $wl^{on}(v) + p^{r*}\tau x^{or}(v)$, where superscript o denotes the firm's offshoring status. Cost minimization, subject to the production technology in Eq. (3), yields the following optimal input demands:

$$l^{on}(v) = \frac{\eta}{\varphi(v)} \left(\frac{\tau p^{r*}}{w}\right)^{1-\eta} q^o(v) \quad \text{and} \quad x^{or}(v) = \frac{(1-\eta)}{\varphi(v)} \left(\frac{w}{\tau p^{r*}}\right)^{\eta} q^o(v). \tag{A.8}$$

This leads to the marginal cost expression:

$$c^o(v) = \frac{(\tau p^{r*})^{1-\eta} w^\eta}{\varphi(v)}. \tag{A.9}$$

To determine p^{r*}, we turn to the host country. Given routine task-specific production technology in Eq. (4), cost minimization for the routine task requires:

$$l^{r*} = (1-\alpha) \left(\frac{t^*}{w^*}\right)^{\alpha} x^{r*} \quad \text{and} \quad e^* = \alpha \left(\frac{w^*}{t^*}\right)^{1-\alpha} x^{r*}. \tag{A.10}$$

Under perfect competition, the price of the offshored input equals its marginal cost:

$$p^{r*} = c^{r*} = (t^*)^\alpha (w^*)^{1-\alpha}. \tag{A.11}$$

Substituting this into Eq. (A.9), together with Eq. (A.6) for domestic firms' marginal costs, yields the marginal cost savings factors presented in Eq. (5) in the main text.

A.3. Proof of Lemma 1

The emissions of a purely domestic firm are given by Eq. (A.5). For an offshoring firm, emissions stem from routine task production abroad. Using Eqs. (A.8) and (A.10) in the identity $\tau x^{or} = x^{r*}$, and substituting the expression for p^{r*} from Eq. (A.11), we obtain:

$$e^o(\varphi) = \tau \frac{(1-\eta)\,\alpha}{\varphi} \left(\frac{w^*}{t^*}\right)^{1-\alpha} \left(\frac{w}{\tau (t^*)^{\alpha} (w^*)^{(1-\alpha)}}\right)^{\eta} q^o(\varphi). \tag{A.12}$$

Dividing both expressions of firm-level emissions – Eqs. (A.5) and (A.12) – by the respective intermediate output $q^d(\varphi)$ and $q^o(\varphi)$ yields firm-level emissions intensities:

$$i^d(\varphi) = \frac{(1-\eta)\alpha}{\varphi} \left(\frac{w}{t}\right)^{1-\alpha} \left[\frac{w}{t^{\alpha} w^{1-\alpha}}\right]^{\eta} \tag{A.13}$$

$$i^o(\varphi) = \tau \frac{(1-\eta)\,\alpha}{\varphi} \left(\frac{w^*}{t^*}\right)^{1-\alpha} \left[\frac{w}{\tau (t^*)^{\alpha} (w^*)^{(1-\alpha)}}\right]^{\eta}. \tag{A.14}$$

Finally, dividing Eq. (A.12) by Eq. (A.5), as well as Eq. (A.14) by Eq. (A.13), yields the emissions and emissions intensity ratios presented in Eq. (7) in the main text.

Using this expression, in Eq. (7), and the definition of κ in Eq. (5), it follows directly that an offshoring firm has a lower emissions intensity than a purely domestic firms, i.e., $i^o(\varphi) < i^d(\varphi)$, if and only if:

$$1 \leq \frac{t}{t^*} < \left(\frac{w}{w^*}\right)^{(1-\alpha)(1-\eta)/[1-\alpha(1-\eta)]} \left(\frac{1}{\tau}\right)^{(1-\eta)/[1-\alpha(1-\eta)]}. \tag{A.15}$$

This completes the proof.

A.4. Derivation of the offshoring equilibrium

A.4.1. Offshoring indifference condition $\mathbf{A}(\chi)$

To derive the offshoring indifference condition $A(\chi)$, we re-arrange the indifference condition for the marginal offshoring firm in Eq. (10) and substitute the occupational choice condition for the marginal manager from Eq. (9):

$$\pi^o(\varphi^o) - \pi^d(\varphi^d) = \pi^d(\varphi^o). \tag{A.16}$$

Dividing both sides by $\pi^d(\varphi^o)$ and applying Eqs. (6) and (11), we obtain:

$$\kappa^{\sigma-1} - \chi^{\frac{\sigma-1}{k}} = 1. \tag{A.17}$$

Solving for κ yields the offshoring indifference condition presented in Eq. (12) of the main text.

A.4.2. Labor market constraint $\mathbf{B}(\tau, \mathbf{t}, \mathbf{t}^*, \chi)$

To derive the labor market constraint $B(\tau, t, t^*, \chi)$, we express the relative wage w/w^* in the definition of κ as a function of χ and model parameters only.

We begin by deriving expressions for aggregate labor income in both countries as a share of total income. This follows from the properties of the CES final goods production function in Eq. (1) and the nested Cobb–Douglas production structure for intermediate goods and routine-task execution, as specified in Eqs. (3) and (4).

Under CES, a constant share $1/\sigma$ of aggregate income accrues to operating profits, and the remaining share $(\sigma-1)/\sigma$ is paid to production factors, in form of labor remuneration and emissions tax payments. In the source country, the Cobb–Douglas production technology implies that a share $(1-\alpha)(1-\eta)$ of production factor payments by purely domestic firms goes to routine task labor, and a share η of production factor payments by both offshoring and purely domestic firms accrues to non-routine task labor. In the host country, labor is employed and remunerated exclusively for the execution of the routine task.

We can thus express aggregate worker income in the two countries as:

$$wL = \frac{\sigma-1}{\sigma} \left[((1-\alpha)(1-\eta)+\eta) \int_{\varphi^d}^{\varphi^o} r^d(\varphi) dG(\varphi) + \eta \int_{\varphi^o}^{\infty} r^o(\varphi) dG(\varphi)\right] N \tag{A.18}$$

$$w^* L^* = \frac{\sigma-1}{\sigma} (1-\alpha)(1-\eta) \int_{\varphi^o}^{\infty} r^o(\varphi) dG(\varphi) N. \tag{A.19}$$

To simplify Eq. (A.18), we use the following elements: $dG(\varphi) = k\varphi^{-(k+1)}$, $r^d(\varphi)/r^d(\varphi^d) = \left(\varphi/\varphi^d\right)^{\sigma-1}$, and $r^o(\varphi)/r^o(\varphi^o) = (\varphi/\varphi^o)^{\sigma-1}$. This yields an expression for the domestic wage rate:

$$w = \gamma^l(\chi) \frac{\sigma-1}{\sigma} \frac{Y}{L} \qquad \text{with} \qquad \gamma^l(\chi) = \frac{1-\alpha(1-\eta)+\eta\chi-(1-\alpha)(1-\eta)\chi^{\frac{k-\sigma+1}{k}}}{1+\chi}, \tag{A.20}$$

which captures the share of production factor payments accruing to labor in the source country.

Analogously, we simplify Eq. (A.19) by using $[1-G(\varphi^o)]/\chi = 1-G(\varphi^d)$, $r^o(\varphi^o) = \sigma\pi^o(\varphi^o)$, $\pi^o(\varphi^o) = \left(1+\chi^{-(\sigma-1)/k}\right)\pi^d(\varphi^d)$, and $Y/M = \sigma\bar{\pi}$, yielding:

$$w^* = \gamma^{l*}(\chi)\frac{\sigma-1}{\sigma}\frac{Y}{L^*} \qquad \text{with} \qquad \gamma^{l*}(\chi) = \frac{(1-\alpha)(1-\eta)\left(\chi+\chi^{\frac{k-\sigma+1}{k}}\right)}{1+\chi}, \tag{A.21}$$

which captures the share of aggregate production factors payments going to workers in the host country.

It follows that $d\gamma^l/d\chi < 0 < d\gamma^{l*}/d\chi$.[52] Hence, as the share of offshoring increases, labor income shifts towards the host country.

Combining Eqs. (A.20) and (A.21) and recalling that $L^* = N^*$, the relative wage becomes:

$$\frac{w}{w^*} = \frac{\gamma^l}{\gamma^{l*}}\frac{N^*}{L}. \tag{A.22}$$

To replace the endogenous variable L, we follow Egger et al. (2015) and use the occupational choice condition for the marginal manager in Eq. (9). The operating profits of the marginal firm are a constant share of average operating profits due to the Pareto distribution of managerial ability:

$$\pi^d(\varphi^d) = \frac{k-\sigma+1}{k}\frac{1}{1+\chi}\bar{\pi} = \frac{k-\sigma+1}{k}\frac{1}{1+\chi}\frac{1}{\sigma}\frac{Y}{M}. \tag{A.23}$$

Substituting the expression for w from Eq. (A.20), we derive

$$\frac{k-\sigma+1}{k}\frac{1}{1+\chi}\frac{1}{\sigma}\frac{Y}{M} = \gamma^l\frac{\sigma-1}{\sigma}\frac{Y}{L}. \tag{A.24}$$

Using the source country's resource constraint in Eq. (13), we solve for the factor allocation:

$$L = \lambda N \qquad \text{and} \qquad M = \frac{1-\lambda}{1+\chi}N \qquad \text{with} \quad \lambda = \left[1+\frac{k-\sigma+1}{\gamma^l k(\sigma-1)}\right]^{-1}, \tag{A.25}$$

where λ denotes the share of workers in the source country's population.

Finally, combining Eqs. (A.22) and (A.25), we express the relative wage rate as:

$$\frac{w}{w^*} = \frac{\gamma^l}{\gamma^{l*}}\frac{N^*}{\lambda N} = \frac{\gamma^l+\frac{k-\sigma+1}{k(\sigma-1)}}{\gamma^{l*}}\frac{N^*}{N}. \tag{A.26}$$

Given that $\gamma^l > \gamma^{l*}$ and $k > \sigma$, it follows that $w > w^*$, provided that N^* is not substantially smaller than N. Moreover, Eq. (A.26) shows that the relative wage w/w^* declines in χ.

This expression leads to the labor market constraint $B(\tau, t, t^*, \chi)$ in Eq. (14) of the main text.

A.4.3. Condition for an interior offshoring equilibrium

An interior equilibrium requires that the share of offshoring firms satisfies $\chi \in (0,1)$. We now derive the necessary conditions on the exogenous parameters that ensure this holds. In particular, cases with $\chi > 1$ must be ruled out, as they violate the occupational choice condition in Eq. (9), which assumes that the marginal firm is purely domestic.

Specifically, an interior offshoring equilibrium requires the upward-sloping offshoring indifference condition $A(\chi)$ and the downward-sloping labor market constraint $B(\tau, t, t^*, \chi)$ to intersect within the interval $\chi \in (0,1)$. A necessary condition is: $\lim_{\chi\to 1} B(\tau, t, t^*, \chi) < A(\chi)$, which provides an upper bound on κ compatible with $\chi < 1$. Solving this yields:

$$\tilde{\kappa} \equiv \left[\frac{1}{\tau}\left(\frac{t}{t^*}\right)^{\alpha}\left(\frac{(k-\sigma+1)+\eta k(\sigma-1)}{(1-\eta)(1-\alpha)k(\sigma-1)}\frac{N^*}{N}\right)^{1-\alpha}\right]^{1-\eta} < 2^{\frac{1}{\sigma-1}}. \tag{A.27}$$

This condition implies that τ must not be too small, and t/t^* must not be too large, highlighting a trade-off in parameter choices. Rearranging yields:

$$\tau > 2^{\frac{1}{(1-\sigma)(1-\eta)}}\left(\frac{t}{t^*}\right)^{\alpha}\left(\frac{(k-\sigma+1)+\eta k(\sigma-1)}{(1-\eta)(1-\alpha)k(\sigma-1)}\frac{N^*}{N}\right)^{1-\alpha} \tag{A.28}$$

and equivalently,

$$\frac{t}{t^*} < 2^{\frac{1}{\alpha(\sigma-1)(1-\eta)}}\tau^{\frac{1}{\alpha}}\left(\frac{(1-\eta)(1-\alpha)k(\sigma-1)}{(k-\sigma+1)+\eta k(\sigma-1)}\frac{N}{N^*}\right)^{\frac{1-\alpha}{\alpha}}. \tag{A.29}$$

[52] For $\chi \to 0$ (autarky), $\gamma^l = 1-\alpha(1-\eta)$ and $\gamma^{l*} = 0$. Approaching the case of all firms offshoring, $\chi \to 1$, $\gamma^l \to \eta$ and $\gamma^{l*} \to (1-\alpha)(1-\eta)$. Consequently, $\gamma^l > \eta > 0.5 > \gamma^{l*}$ given our parameter restrictions.

Appendix B. Unilateral environmental policy reform

B.1. Proof of Proposition 1

B.1.1. Effect on the share of offshoring firms

We apply the implicit function theorem to derive the change in χ in response to a rise in t. To this end, we make use of Eqs. (12) and (14), the two links between κ and χ, and define the implicit function:

$$F(\chi,\tau,t,t^*) \equiv B(\chi,\tau,t,t^*) - A(\chi) = 0 \tag{B.1}$$
$$= \left[\frac{1}{\tau}\left(\frac{t}{t^*}\right)^{\alpha}\left(\frac{\gamma^l}{\gamma^{l*}}\frac{N^*}{\lambda N}\right)^{1-\alpha}\right]^{1-\eta} - \left(1+\chi^{\frac{\sigma-1}{k}}\right)^{\frac{1}{\sigma-1}} = 0.$$

Applying the implicit function theorem, we get $d\chi/dt = -(\partial F/\partial t)/(\partial F/\partial \chi) > 0$ with

$$\frac{\partial F}{\partial t} = \alpha(1-\eta)\left(\frac{t}{t^*}\right)^{\alpha(1-\eta)-1}\frac{1}{t^*}\left[\frac{1}{\tau}\left(\frac{\gamma^l}{\gamma^{l*}}\frac{N^*}{\lambda N}\right)^{1-\alpha}\right]^{1-\eta} > 0 \tag{B.2}$$

and

$$\frac{\partial F}{\partial \chi} = (1-\alpha)(1-\eta)\left[\frac{1}{\tau}\left(\frac{t}{t^*}\right)^{\alpha}\left(\frac{N^*}{N}\right)^{1-\alpha}\right]^{1-\eta}\left(\frac{\gamma^l}{\gamma^{l*}\lambda}\right)^{-\eta(1-\alpha)-\alpha} \tag{B.3}$$
$$\times \frac{\frac{d\gamma^l}{d\chi}\gamma^{l*} - \frac{d\gamma^{l*}}{d\chi}\left[\gamma^l + \frac{k-\sigma+1}{k(\sigma-1)}\right]}{\left(\gamma^{l*}\right)^2} - \frac{1}{k}\left(1+\chi^{\frac{\sigma-1}{k}}\right)^{\frac{2-\sigma}{\sigma-1}}\chi^{\frac{\sigma-1-k}{k}} < 0,$$

where the negative sign follows from $d\gamma^l/d\chi < 0$ and $d\gamma^{l*}/d\chi > 0$.

These results confirm the first part – regarding the increase in the share of offshoring firms – of Proposition 1, part (i).

B.1.2. Effect on the factor allocation and labor decomposition

We note that t does not directly affect L and M, see the expressions in Eq. (A.25); all effects run indirectly via χ. Accounting for this, we derive the emissions tax-induced changes in the factor allocation as follows:

$$\frac{dL}{dt} < 0, \qquad \frac{d(1+\chi)M}{dt} > 0, \qquad \frac{d\chi M}{dt} > 0, \qquad \frac{dM}{dt} \gtreqless 0, \qquad \frac{d(1-\chi)M}{dt} \gtreqless 0. \tag{B.4}$$

The first expression follows from Eq. (A.25) and $d\lambda/d\chi < 0$. The second expression is a direct implication, taking Eq. (13) into account. The third expression is again directly implied by Eq. (A.25) and $d\lambda/d\chi < 0$.[53] The fourth expression shows an ambiguous effect on M. We can (explicitly) solve for a threshold level of χ:

$$\tilde{\chi} = \left[\frac{(k-\sigma+1)(\sigma-1)(1-\eta)(1-\alpha)}{k-\sigma+1+k(\sigma-1)\eta}\right]^{\frac{k}{\sigma-1}} > 0, \tag{B.5}$$

with $dM/dt > 0$ for $\chi < \tilde{\chi}$ and $dM/dt < 0$ for $\chi > \tilde{\chi}$ (cf. Egger et al., 2015). Finally, the fifth expression is a combination of the third and fourth expression as $(1-\chi)M = M - \chi M$. To specify the sign of the tax effect, we (implicitly) solve for a second threshold level of χ:

$$\tilde{\chi}^d = \left[\frac{[(k-\sigma+1)+\chi(\sigma-1)](\sigma-1)(1-\eta)(1-\alpha)}{2[k-\sigma+1+k(\sigma-1)\eta]+k(\sigma-1)(1-\alpha)(1-\eta)}\right]^{\frac{k}{\sigma-1}} > 0, \tag{B.6}$$

where it is straightforward that $\tilde{\chi}^d < \tilde{\chi}$. Accordingly, for $\chi < \tilde{\chi}^d$ it holds that $d(1-\chi)M/dt > 0$, while $d(1-\chi)M/dt < 0$ for $\chi > \tilde{\chi}^d$.

We can decompose source country labor with regard to task and firm type to derive the two insights discussed in the main text.

First, we decompose source country labor into labor employed for the non-routine and the routine task, i.e., $L = L^n + L^{dr}$. All workers performing the non-routine task are necessarily based in the source country, no matter the firm type that employs them: $L^n = L^{dn} + L^{on}$. Equivalent to Eqs. (A.18) and (A.19), we can derive aggregate non-routine task labor from shares in aggregate income. This gives the following expressions:

$$L^n = \frac{\eta k(\sigma-1)N}{k-\sigma+1+\gamma^l k(\sigma-1)}, \qquad L^{dn} = L^n\frac{1-\chi^{\frac{k-\sigma+1}{k}}}{1+\chi}, \qquad L^{on} = L^n\frac{\chi^{\frac{k-\sigma+1}{k}}+\chi}{1+\chi}. \tag{B.7}$$

Since $d\gamma^l/d\chi < 0$, it follows that aggregate non-routine task labor increases in t, $dL^n/dt > 0$. Moreover, we find an increase in aggregate employment of non-routine task labor of offshoring firms, while it decreases for purely domestic firms. The increasing effect dominates such that $dL^{on}/dt > -dL^{dn}/dt > 0$. Routine task, in turn, is only conducted in the source country by purely domestic firms. Since $dL^n/dt > 0$ and $dL/dt < 0$, it must hold that $dL^{dr}/dt < 0$.

[53] Given the Pareto distribution of firm productivity, the share of offshoring firms, χ, and the total mass of active firms, M, determine the cutoff productivity of the marginal offshoring firm, φ^o, via $\chi M = (\varphi^o)^{-k} N$. Accordingly, an increase in χ implies a decline in φ^o.

Second, we decompose source country labor into labor employed by purely domestic firms and offshoring firms, i.e., $L = L^d + L^{on}$. We derived in the first decomposition of this Appendix that purely domestic firms reduce both non-routine and routine task labor. By contrast, offshoring firms increase their non-routine task labor. Hence, $dL^d/dt < 0$, while $dL^{on}/dt > 0$.

Jointly, these results can be summarized as in Eq. (15) in the main text. They confirm the second statement in part (i) – regarding the increase in the mass of offshoring firms – and part (ii) of Proposition 1.

B.1.3. Effect on wage rates

As shown in Appendix A.4.2, both wage rates are determined by global aggregate output Y, the income shares of source and host country workers (γ^l and γ^{l*}), and the aggregate number of workers in the respective country (L and L^*):

$$w = \gamma^l(\chi)\frac{\sigma-1}{\sigma}\frac{Y}{L} \qquad \text{and} \qquad w^* = \gamma^{l*}(\chi)\frac{\sigma-1}{\sigma}\frac{Y}{L^*}. \tag{B.8}$$

Offshoring shifts labor income from the source to the host country workers, expressed in $d\gamma^l/d\chi < 0 < d\gamma^{l*}/d\chi$ and it decreases labor demand, hence, the mass of workers in the source country, $dL/d\chi < 0$, while the mass of workers in the host country is exogenous by assumption and does not change. From Eq. (A.25) it is straightforward that combining γ^l with L leads to a term which decreases in χ, hence, also in t. We know that the share of income going to host country workers is increasing in t, which means, that we have a positive partial effect here, and a negative one for the source wage rate.

To understand the entire effect of t on both wage rates, it is essential to express Y as a function of χ and parameters only, to derive the response to an increase in t.

To express aggregate output in closed form, we start by linking it to expressions of the marginal firm, beginning with Eq. (2):

$$q^d(\varphi^d) = Yp(\varphi^d)^{-\sigma}. \tag{B.9}$$

Furthermore, we can express the price set by marginal firm, knowing constant mark-up pricing and the marginal costs of the marginal firm from Eq. (A.6):

$$p(\varphi^d) = \frac{\sigma}{\sigma-1}\frac{\left(t^\alpha w^{1-\alpha}\right)^{1-\eta} w^\eta}{\varphi^d}. \tag{B.10}$$

Next, we replace source country wage rate using Eq. (A.24):

$$p(\varphi^d) = \frac{\sigma}{\sigma-1}\frac{t^\alpha}{\varphi^d}\left(\frac{k-\sigma+1}{k}\frac{1}{1+\chi}\frac{1}{\sigma}\frac{Y}{M}\right)^{1-\alpha(1-\eta)}. \tag{B.11}$$

We plug this solution into Eq. (B.9) and use $q^d(\varphi^d) = (\sigma-1)\varphi^d(w/t)^{\alpha(1-\eta)}$ as output of the marginal firm. Solving this for Y gives us:

$$Y = \sigma\left[M(1+\chi)\frac{k}{k-\sigma+1}\right]^{\frac{(\sigma-1)(1-\alpha(1-\eta))+1}{(\sigma-1)(1-\alpha(1-\eta))}}\left[\frac{1}{t}\right]^{\frac{\alpha(1-\eta)}{1-\alpha(1-\eta)}}\left[\frac{\sigma-1}{\sigma}\varphi^d\right]^{\frac{1}{1-\alpha(1-\eta)}}. \tag{B.12}$$

Using our closed-form expressions of M from Eq. (A.25), along with $\varphi^d = (N/M)^{1/k}$, we end at:

$$\begin{aligned} Y = {} & \left[\frac{1}{k-\sigma+1+\gamma^l k(\sigma-1)}\right]^{\frac{k(\sigma-1)(1-\alpha(1-\eta))+k-\sigma+1}{k(\sigma-1)(1-\alpha(1-\eta))}}[Nk]^{\frac{(\sigma-1)(1-\alpha(1-\eta))+1}{(\sigma-1)(1-\alpha(1-\eta))}}\left(\frac{1}{t\sigma}\right)^{\frac{\alpha(1-\eta)}{1-\alpha(1-\eta)}} \\ & \times\left(\left(\frac{1+\chi}{k-\sigma+1}\right)^{\frac{1}{k}}(\sigma-1)\right)^{\frac{1}{1-\alpha(1-\eta)}}. \end{aligned} \tag{B.13}$$

The total effect of t on Y is calculated as: $dY/dt = \partial Y/\partial t + \partial Y/\partial\chi \times d\chi/dt$. For the direct effect of t, we get:

$$\frac{\partial Y}{\partial t} = -\frac{\alpha(1-\eta)}{1-\alpha(1-\eta)}\frac{1}{t}Y < 0. \tag{B.14}$$

For the direct effect of χ, we get:

$$\begin{aligned} \frac{\partial Y}{\partial\chi} = {} & Y\times\left[\frac{[k(\sigma-1)(1-\alpha(1-\eta))+(k-\sigma+1)](k-\sigma+1)(1-\alpha)(1-\eta)\chi^{\frac{k-\sigma+1}{k}-1}}{k(1+\chi)(1-\alpha(1-\eta))[k-\sigma+1+\gamma^l k(\sigma-1)]}\right. \\ & \left.+\frac{[k(\sigma-1)(1-\alpha(1-\eta))+(k-\sigma+1)]k(\gamma^l-\eta)+k-\sigma+1+\gamma^l k(\sigma-1)}{k(1+\chi)(1-\alpha(1-\eta))[k-\sigma+1+\gamma^l k(\sigma-1)]}\right] > 0. \end{aligned} \tag{B.15}$$

As expected, offshoring increases aggregate output due to marginal cost savings, while an emissions tax increase makes production more costly, hence decreases aggregate output. Put together and extracting Y, we have for the total effect:

$$\begin{aligned} \frac{dY}{dt} = {} & Y\times\left[-\frac{\alpha(1-\eta)}{1-\alpha(1-\eta)}\frac{1}{t}\right. \\ & \left.+\left[\frac{[k(\sigma-1)(1-\alpha(1-\eta))+(k-\sigma+1)](k-\sigma+1)(1-\alpha)(1-\eta)\chi^{\frac{k-\sigma+1}{k}-1}}{k(1+\chi)(1-\alpha(1-\eta))[k-\sigma+1+\gamma^l k(\sigma-1)]}\right.\right. \end{aligned}$$

$$+\frac{[k(\sigma-1)(1-\alpha(1-\eta))+(k-\sigma+1)]k(\gamma^l-\eta)+k-\sigma+1+\gamma^l k(\sigma-1)}{k(1+\chi)(1-\alpha(1-\eta))[k-\sigma+1+\gamma^l k(\sigma-1)]}\Bigg]\times\frac{d\chi}{dt}\Bigg]. \tag{B.16}$$

To show the sign of the total effect we make use of $d\chi/dt$ with Eqs. (B.2) and (B.3) and set Eq. (B.16) equal to zero. Rearranging (while ignoring Y) gives us:

$$\begin{aligned}
&\Bigg\{\Bigg[\frac{[k(\sigma-1)(1-\alpha(1-\eta))+(k-\sigma+1)](k-\sigma+1)(1-\alpha)(1-\eta)\chi^{\frac{k-\sigma+1}{k}-1}}{k(1+\chi)\left(\chi+\chi^{\frac{k-\sigma+1}{k}}\right)[k-\sigma+1+\gamma^l k(\sigma-1)]}\\
&+\frac{[k(\sigma-1)(1-\alpha(1-\eta))+(k-\sigma+1)]k(\gamma^l-\eta)+k-\sigma+1+\gamma^l k(\sigma-1)}{k(1+\chi)\left(\chi+\chi^{\frac{k-\sigma+1}{k}}\right)[k-\sigma+1+\gamma^l k(\sigma-1)]}\Bigg]\left(\chi+\chi^{\frac{k-\sigma+1}{k}}\right)\\
&-\Bigg[\frac{(1+\chi)[(k-\sigma+1)(1-\alpha)(1-\eta)\chi^{\frac{k-\sigma+1}{k}-1}+k(\gamma^l-\eta)]}{(1+\chi)k[k-\sigma+1+\gamma^l k(\sigma-1)]\left(\chi^{\frac{k-\sigma+1}{k}}+\chi\right)}\\
&\times\frac{[(1-\alpha(1-\eta))k(\sigma-1)+k-\sigma+1]}{(1+\chi)k[k-\sigma+1+\gamma^l k(\sigma-1)]\left(\chi^{\frac{k-\sigma+1}{k}}+\chi\right)}\Bigg]\Bigg\}\times B(\chi,\tau,t,t^*)\\
&\gtrless\frac{1}{k}\times A(\chi)\times\frac{1}{\chi^{\frac{k-\sigma+1}{k}}+\chi},
\end{aligned} \tag{B.17}$$

with $A(\chi)$ and $B(\chi,\tau,t,t^*)$ stemming from Eq. (B.1). While the first two rows are positive, the third and fourth are negative. It is straightforward to see that the third and fourth row are greater than the first and second row because of $(1+\chi)>(\chi+\chi^{(k-\sigma+1)/k})$, while ignoring $k-\sigma+1+\gamma^l k(\sigma-1)$ in the second row. Focusing on this part, we compare it with the right-hand side term. Since in equilibrium both $\kappa-\chi$ conditions intersect, we can cancel out these two terms. So we end for our comparison at:

$$\frac{(k-\sigma+1+\gamma^l k(\sigma-1))\left(\chi^{\frac{k-\sigma+1}{k}}+\chi\right)}{k(1+\chi)(\chi^{\frac{k-\sigma+1}{k}}+\chi)[k-\sigma+1+\gamma^l k(\sigma-1)]}\gtrless\frac{1}{k}\frac{1}{\chi^{\frac{k-\sigma+1}{k}}+\chi}. \tag{B.18}$$

Canceling out and rearranging leads to:

$$1\gtrless\frac{1+\chi}{\chi+\chi^{\frac{k-\sigma+1}{k}}}. \tag{B.19}$$

The right-hand side is decreasing in χ and reaches one when χ approaches its maximum level of one. That means, for all $\chi\in(0,1)$ the left-hand side is smaller than the right-hand side, which means the negative term dominates the positive term. Hence, we have shown that $dY/dt<0$.

Together with the result that less income is going to source country workers, it is clear that the source country wage rate decreases in the source country emissions tax, $dw/dt<0$.

To calculate the total effect of the tax rate increase on host country wage, we follow the same procedure as for the aggregate output.

The direct effect of t is given by:

$$\frac{\partial w^*}{\partial t}=-\frac{\sigma-1}{\sigma}Y\left(\gamma^{l*}\frac{\alpha(1-\eta)}{1-\alpha(1-\eta)}\frac{1}{t}\right)<0. \tag{B.20}$$

The effect of offshoring is given by:

$$\begin{aligned}
\frac{dw^*}{d\chi}=&\frac{\sigma-1}{\sigma}Y\Bigg[\gamma^{l*}\frac{k-\sigma+1+\gamma^l k(\sigma-1)}{k(1+\chi)(1-\alpha(1-\eta))[k-\sigma+1+\gamma^l k(\sigma-1)]}\\
&+\left[k(\sigma-1)(1-\alpha(1-\eta))^2+(k-\sigma+1)((1-\alpha(1-\eta))+\gamma^{l*})\right]\\
&\times\frac{\left[(k-\sigma+1)(1-\alpha)(1-\eta)\chi^{-\frac{\sigma-1}{k}}+k(\gamma^l-\eta)\right]}{k(1+\chi)(1-\alpha(1-\eta))[k-\sigma+1+\gamma^l k(\sigma-1)]}\Bigg]>0.
\end{aligned} \tag{B.21}$$

Consequently, we can write for the total effect:

$$\begin{aligned}
\frac{dw^*}{dt}=&\frac{\sigma-1}{\sigma}Y\Bigg[-\gamma^{l*}\frac{\alpha(1-\eta)}{1-\alpha(1-\eta)}\frac{1}{t}+\Bigg[\gamma^{l*}\frac{k-\sigma+1+\gamma^l k(\sigma-1)}{k(1+\chi)(1-\alpha(1-\eta))[k-\sigma+1+\gamma^l k(\sigma-1)]}\\
&+\left[k(\sigma-1)(1-\alpha(1-\eta))^2+(k-\sigma+1)((1-\alpha(1-\eta))+\gamma^{l*})\right]\\
&\times\frac{\left[(k-\sigma+1)(1-\alpha)(1-\eta)\chi^{-\frac{\sigma-1}{k}}+k(\gamma^l-\eta)\right]}{k(1+\chi)(1-\alpha(1-\eta))[k-\sigma+1+\gamma^l k(\sigma-1)]}\Bigg]\times\frac{d\chi}{dt}\Bigg].
\end{aligned} \tag{B.22}$$

We use Eq. (B.1) for $d\chi/dt$ and, again, compare the negative direct effect with the positive offshoring effect. Tedious calculations and the simplification in equilibrium $A(\chi) = B(\chi, t, t^*, \tau)$ lead us to:

$$
\begin{aligned}
&- \frac{\left(1-\chi^{\frac{k-\sigma+1}{k}}\right)\gamma^{l*}\left[k-\sigma+1+\gamma^l k(\sigma-1)\right]}{k(1+\chi)[k-\sigma+1+\gamma^l k(\sigma-1)]\left(\chi+\chi^{\frac{k-\sigma+1}{k}}\right)} \\
&+ \frac{\left[k(\sigma-1)[\eta-\alpha(1-\eta)(1-\alpha(1-\eta))]+(k-\sigma+1)\frac{(1-\alpha(1-\eta))\chi+\chi^{\frac{k-\sigma+1}{k}}(1-\alpha)(1-\eta)+\eta}{1+\chi}\right]}{k(1+\chi)[k-\sigma+1+\gamma^l k(\sigma-1)]} \\
&\times \left[(k-\sigma+1)(1-\alpha)(1-\eta)\chi^{-\frac{\sigma-1}{k}}+k(\gamma^l-\eta)\right] > 0. \qquad \text{(B.23)}
\end{aligned}
$$

Comparing the parts $k-\sigma+1$ as well as $k(\sigma-1)$ separately with each other, together with our assumptions $\eta > 0,5$, $k > \sigma$ and $k > 1$, reveals that $dw^*/dt > 0$. Accordingly, the relative wage declines: $d(w/w^*)/dt < 0$.

These results confirm part (iii) of Proposition 1.

B.2. Proof of Proposition 2

B.2.1. Effect on source country's aggregate emissions

To derive aggregate source country's emissions as given by Eq. (16) in the main text, we follow the same strategy based on income shares as in Appendix A.4. Unlike labor income, the revenue share linked to emissions tax income originates only from routine task production – and only purely domestic firms execute this part of production in the source country:

$$tE = \frac{\sigma-1}{\sigma}\alpha(1-\eta)\int_{\varphi^d}^{\varphi^o} r^d(\varphi)dG(\varphi)N. \qquad \text{(B.24)}$$

Reformulating yields Eq. (16) in the main text with

$$\gamma^e(\chi) \equiv \frac{\alpha(1-\eta)(1-\chi^{\frac{k-\sigma+1}{k}})}{1+\chi}. \qquad \text{(B.25)}$$

It follows directly that $d\gamma^e/dt = d\gamma^e/d\chi \cdot d\chi/dt < 0$, since the numerator decreases and the denominator increases in χ. This effect amplifies the obvious negative direct effect of the source country emissions tax as local emissions-intensive production becomes more costly.

Combined with the result from Appendix B.1.3 that aggregate income falls with the tax, $dY/dt < 0$, it follows that aggregate source country emissions must also fall in t, i.e., $dE/dt < 0$.

This result confirms part (i) of Proposition 2.

B.2.2. Effect on host country's aggregate emissions

Host country aggregate emissions can be derived analogously to those in the source country. Starting from

$$t^*E^* = \frac{\sigma-1}{\sigma}\alpha(1-\eta)\int_{\varphi^o}^{\infty} r^o(\varphi)dG(\varphi)N, \qquad \text{(B.26)}$$

and following the same steps as in Eq. (16), we obtain:

$$E^* = \gamma^{e*}(\chi)\frac{\sigma-1}{\sigma}\frac{Y}{t^*} \qquad \text{with} \quad \gamma^{e*}(\chi) \equiv \frac{\alpha(1-\eta)(\chi+\chi^{\frac{k-\sigma+1}{k}})}{1+\chi}, \qquad \text{(B.27)}$$

where $\gamma^{e*}(\chi)$ denotes the share of aggregate income net of aggregate operating profits associated with emissions tax payments in the host country. Since $\gamma^{e*}(\chi)$ is increasing in χ, it opposes the tax-induced decline in Y.

To establish the tax effect on E^*, we exploit the simplified structure of the host country, which only produces the routine task using Cobb–Douglas technology. Hence, we can relate emissions tax revenue to wage income:

$$w^*L^* = \gamma^{l*}\frac{\sigma-1}{\sigma}Y = (1-\alpha)I^*, \qquad t^*E^* = \gamma^{e*}\frac{\sigma-1}{\sigma}Y = \alpha I^*, \qquad \text{(B.28)}$$

which implies:

$$E^* = \frac{\alpha}{1-\alpha}\frac{w^*}{t^*}N^*. \qquad \text{(B.29)}$$

Since N^* and t^* are constant, it follows that $dE^* \propto dw^*$. By Proposition 1, part (iii), the host country wage w^* rises with t, confirming that E^* increases as well. This result confirms part (ii) of Proposition 2.

B.2.3. Effect on global emissions

Making use of Eqs. (16), (B.25), and (B.27), we derive global emissions as

$$E^W = \left(\frac{\gamma^e}{t} + \frac{\gamma^{e*}}{t^*}\right)\frac{\sigma-1}{\sigma}Y = \frac{1}{t}\left(\gamma^e + \gamma^{e*}\frac{t}{t^*}\right)\frac{\sigma-1}{\sigma}Y$$
$$= \alpha(1-\eta)\frac{\sigma-1}{\sigma}\left(\frac{1-\chi^{\frac{k-\sigma+1}{k}}}{1+\chi}\frac{1}{t} + \frac{\chi+\chi^{\frac{k-\sigma+1}{k}}}{1+\chi}\frac{1}{t^*}\right)Y, \tag{B.30}$$

which simplifies to Eq. (17) in the main text. This formulation highlights that the share of global emissions generated in the source country declines as either the emissions tax t increases or the share of offshoring firms χ rises. Since both effects reduce the ratio E/E^*, emissions increasingly shift to the host country as t rises.

In the limit, as $t \to \infty$, source country emissions $E \to 0$, while host country emissions $E^* \to \infty$ due to rising w^*. As a result, the emission ratio $E/E^* \to 0$, implying $E^W \to E^* \to \infty$. Conversely, for low t – given τ and other fundamentals – the offshoring share χ is small and global emissions initially fall, driven by the decline in E, which carries more weight in the emission aggregate. This confirms the non-monotonic behavior of E^W in t – global emissions decline at first and rise beyond a threshold tax level, exhibiting a U-shaped response to the unilateral tax reform.

This pattern depends critically on whether the offshoring equilibrium remains interior, i.e., $\chi \in (0,1)$ for all t. Since κ, and hence χ increases in t for fixed τ, trade costs must be large enough to ensure that χ never reaches or exceeds one – see the conditions derived in Appendix A.4.3. Accordingly, under valid parameter constellations, in particular a sufficiently high τ, the global emissions response E^W exhibits the claimed U-shape.

An increase in τ raises the threshold tax differential at which global emissions reach their minimum and begin to rise. This is because χ increases in the tax differential t/t^* but decreases in trade costs τ – so trade frictions dampen the expansion of offshoring and thereby delay the shift of emissions to the host country.

These results confirm part (iii) of Proposition 2.

B.3. Derivations for emissions decompositions

B.3.1. Derivations for decomposition i in Section 3.3.1

For Decomposition I and its focus on labor reallocation across tasks, we first express aggregate emissions in the source country in terms of domestic labor allocated to the routine task, L^{dr}. From the Cobb–Douglas routine task production technology given in Eq. (4) and aggregated across firms, it follows that $\alpha t E = (1-\alpha) w L^{dr}$. Rearranged, this yields:[54]

$$E = \frac{\alpha}{1-\alpha}\frac{w}{t}L^{dr}. \tag{B.31}$$

To arrive at Eq. (18) in the main text, we multiply and divide by total labor L, intermediate output $Q \equiv \int_{\varphi^d}^{\infty} q(\varphi)\,d\varphi$, and final goods output Y.

B.3.2. Derivations for decomposition II in Section 3.3.2

This section derives the components of global emissions E^W used in Decomposition II, as detailed in Eq. (19) in the main text. While Y, ψ, and M are defined above, we here focus on the firm-level averages in output and emissions intensity, $\bar{q}$ and $\bar{\imath}$. We also characterize how these variables respond to a unilateral increase in t where possible, complementing our earlier derivations for effects on M and Y. Numerical simulations are presented in Section 3.3.2 of the main text and in Online Appendix Y.3.

The averages across all firms in output and emissions intensity, $\bar{q}$ and $\bar{\imath}$, are each defined as a combination of firm-type specific averages:

$$\bar{q} = (1-\chi)\cdot\bar{q}^d + \chi\cdot\bar{q}^o \quad \text{and} \quad \bar{\imath} = (1-\chi)\cdot\frac{\bar{q}^d}{\bar{q}}\cdot\bar{\imath}^d + \chi\cdot\frac{\bar{q}^o}{\bar{q}}\cdot\bar{\imath}^o. \tag{B.32}$$

Accordingly, four key components – $\bar{q}^d$, $\bar{q}^o$, $\bar{\imath}^d$, and $\bar{\imath}^o$ – govern the evolution of $\bar{q}$ and $\bar{\imath}$. We derive each in turn below.

First, we derive the average output of purely domestic firms, $\bar{q}^d$, defined as:

$$\bar{q}^d = \int_{\varphi^d}^{\varphi^o} q^d(\varphi)\frac{g(\varphi)}{G(\varphi^o)-G(\varphi^d)}d\varphi. \tag{B.33}$$

Following the steps in Appendix A.4.2 and using $q^d(\varphi^d) = (\sigma-1)\varphi^d(w/t)^{\alpha(1-\eta)}$ – inferred from the occupational choice indifference condition, Eq. (9) – this yields:

$$\bar{q}^d = \frac{k}{k-\sigma}\frac{1-\chi^{\frac{k-\sigma}{k}}}{1-\chi}q^d(\varphi^d) = \frac{k(\sigma-1)}{k-\sigma}\frac{1-\chi^{\frac{k-\sigma}{k}}}{1-\chi}\left(\frac{w}{t}\right)^{\alpha(1-\eta)}\varphi^d. \tag{B.34}$$

[54] Alternatively, one arrives at this expression by substituting for γ^e in the income-based expression for aggregate emissions in the source country, given in Eq. (16), using the identity $\gamma^e = (\alpha\sigma w L^{dr})/[(1-\alpha)(\sigma-1)Y]$, which follows from $\gamma^e/\gamma^l L = \alpha/(1-\alpha)L^{dr}$ and Eq. (A.20).

Next, we replace the marginal productivity level φ^d by the output-weighted average productivity of purely domestic firms, defined as:

$$\frac{1}{\bar{\varphi}^d} = \int_{\varphi^d}^{\varphi^o} \frac{q^d(\varphi)}{\bar{q}^d} \frac{1}{\varphi} \frac{g(\varphi)}{G(\varphi^o) - G(\varphi^d)} d\varphi. \tag{B.35}$$

Solving the integral yields:

$$\frac{1}{\bar{\varphi}^d} = \frac{k}{k-\sigma+1} \frac{1-\chi^{\frac{k-\sigma+1}{k}}}{1-\chi} \frac{q^d(\varphi^d)}{\bar{q}^d} \frac{1}{\varphi^d}. \tag{B.36}$$

Substituting for $q^d(\varphi^d)/\bar{q}^d$ from Eq. (B.34), first term, gives:

$$\frac{1}{\bar{\varphi}^d} = \frac{k-\sigma}{k-\sigma+1} \frac{1-\chi^{\frac{k-\sigma+1}{k}}}{1-\chi^{\frac{k-\sigma}{k}}} \frac{1}{\varphi^d}. \tag{B.37}$$

Inserting this result into Eq. (B.34), second term, we obtain the final expression for the average output of purely domestic firms:

$$\bar{q}^d = \frac{k(\sigma-1)}{k-\sigma+1} \frac{1-\chi^{\frac{k-\sigma+1}{k}}}{1-\chi} \left(\frac{w}{t}\right)^{\alpha(1-\eta)} \bar{\varphi}^d. \tag{B.38}$$

Second, we derive the average output level of offshoring firms, $\bar{q}^o$, defined as:

$$\bar{q}^o = \int_{\varphi^o}^{\infty} q^o(\varphi) \frac{g(\varphi)}{1-G(\varphi^o)} d\varphi. \tag{B.39}$$

Solving the integral, using $q^o(\varphi) = q^d(\varphi)\kappa^\sigma$ and Eq. (12), we obtain:

$$\bar{q}^o = \frac{k}{k-\sigma} \frac{(1+\chi^{\frac{\sigma-1}{k}})^{\frac{\sigma}{\sigma-1}}}{\chi^{\frac{\sigma}{k}}} q^d(\varphi^d). \tag{B.40}$$

Next, we express $q^d(\varphi^d)$ in terms of the weighted average productivity level of offshoring firms, defined as:

$$\frac{1}{\bar{\varphi}^o} = \int_{\varphi^o}^{\infty} \frac{q^o(\varphi)}{\bar{q}^o} \frac{1}{\varphi} \frac{g(\varphi)}{1-G(\varphi^o)} d\varphi. \tag{B.41}$$

Solving the integral and using Eq. (B.40), yields:

$$\frac{1}{\bar{\varphi}^o} = \frac{k-\sigma}{k-\sigma+1} \frac{1}{\varphi^o}. \tag{B.42}$$

To express $\bar{q}^o$ in terms of $\bar{\varphi}^o$, we use $q^d(\varphi^d) = (\sigma-1)\varphi^d(w/t)^{\alpha(1-\eta)}$ – from Eq. (9) – and the relation $\varphi^d = \varphi^o \cdot \chi^{1/k}$ – from the definition of χ in Eq. (11). Substituting into Eq. (B.40) and using Eq. (B.42), we obtain the final expression for the average output of offshoring firms:

$$\bar{q}^o = \frac{k(\sigma-1)}{k-\sigma+1} \frac{\left(1+\chi^{\frac{\sigma-1}{k}}\right)^{\frac{\sigma}{\sigma-1}}}{\chi^{\frac{\sigma-1}{k}}} \left(\frac{w}{t}\right)^{\alpha(1-\eta)} \bar{\varphi}^o. \tag{B.43}$$

Finally, we can express average emissions intensities by firm type using Eqs. (A.13) and (A.14), evaluated at the average productivity levels derived above:

$$\bar{i}^d = \alpha \left(\frac{t}{w}\right)^{-(1-\alpha)} \cdot (1-\eta) \left(\frac{t}{w}\right)^{-\alpha\eta} \cdot \frac{1}{\bar{\varphi}^d}, \tag{B.44}$$

$$\bar{i}^o = \tau \cdot \alpha \left(\frac{t^*}{w^*}\right)^{-(1-\alpha)} \cdot (1-\eta) \left[\frac{w}{\tau (w^*)^{1-\alpha} (t^*)^{\alpha}}\right]^{\eta} \cdot \frac{1}{\bar{\varphi}^o}. \tag{B.45}$$

We first examine the effect of a unilateral tax increase on average output of purely domestic firms. Starting from Eq. (B.34), and following the procedure from Appendix B.1.3, we use $\varphi^d = [(1+\chi)/(1-\lambda)]^{(1/k)}$ to obtain:

$$\begin{aligned}\bar{q}^d = &\frac{k(\sigma-1)}{k-\sigma} \frac{1-\chi^{\frac{k-\sigma}{k}}}{1-\chi} \left[\frac{1}{k-\sigma+1+\gamma' k(\sigma-1)}\right]^{\frac{\alpha(1-\eta)k-(\sigma-1)}{k(\sigma-1)(1-\alpha(1-\eta))}} [Nk]^{\frac{1}{(\sigma-1)(1-\alpha(1-\eta))}} \\ &\times \left(\frac{\sigma-1}{\sigma}\right)^{\frac{1}{1-\alpha(1-\eta)}} \left(\frac{1}{t}\right)^{\frac{\alpha(1-\eta)}{1-\alpha(1-\eta)}} (1+\chi)^{\frac{1}{k(1-\alpha(1-\eta))}} \left[\frac{1}{k-\sigma+1}\right]^{\frac{1}{k}}.\end{aligned} \tag{B.46}$$

This expression closely resembles the solution for aggregate output Y in Eq. (B.13), with two notable differences. First, the χ-term in the first row. It is strictly decreasing in χ, and thus in t. Second, the exponent of the first bracketed term – responsible for the positive χ-effect on aggregate output – is smaller here (or may even be negative, depending on parameter values). Since the positive χ-effect is already dominated in the case of Y by the negative direct tax effect, the same conclusion must hold more strongly for $\bar{q}^d$. We therefore conclude that the average output of purely domestic firms declines with the emissions tax rate, i.e., $d\bar{q}^d/dt < 0$.

While the effect of a unilateral increase in t on average emissions intensity of purely domestic firms is a priori ambiguous – due to its unclear effect on average productivity, in line with part (ii) of Proposition 1 – we can derive clear results for offshoring firms. We first show that $d\bar{i}^o/dt > 0$, followed by $d\bar{q}^o/dt < 0$.

Using Eq. (B.42), the average emissions intensity of offshoring firms can be expressed as a multiple of the marginal offshoring firm's intensity:

$$\bar{i}^o = i^o(\varphi^o)\frac{k-\sigma}{k-\sigma+1}. \tag{B.47}$$

Using the relations $i^o(\varphi) = i^d(\varphi)t/t^*\kappa^{-1}, i^d(\varphi) = i^d(\varphi^d)(\varphi^d/\varphi)$, and $\chi^{(1/k)}/\varphi^d = 1/\varphi^o$, together with Eq. (A.13) for the emissions intensity of the marginal domestic firm, we obtain:

$$i^o(\varphi^o) = \alpha(1-\eta)\frac{\sigma-1}{\sigma}\frac{1}{t}\left(\frac{k}{k-\sigma+1+\gamma^l k(\sigma-1)}N\right)^{\frac{1}{\sigma-1}}\frac{t}{t^*}\kappa^{-1}\left[\kappa^{\sigma-1}-1\right]^{\frac{1}{\sigma-1}}. \tag{B.48}$$

This simplifies to:

$$i^o(\varphi^o)^{\sigma-1} = \left[\alpha(1-\eta)\frac{\sigma-1}{\sigma}\frac{1}{t^*}\right]^{\sigma-1}\left(\frac{k}{k-\sigma+1+\gamma^l k(\sigma-1)}N\right)\left[1-\frac{1}{\kappa^{\sigma-1}}\right]. \tag{B.49}$$

Since both of the final terms in this expression – through γ^l and κ – increase in χ, we obtain $di^o(\varphi^o)/dt > 0$. It then follows from Eq. (B.47) that $d\bar{i}^o/dt > 0$ as well.[55]

To determine the effect on average output, we exploit the identity $\bar{e}^o = \bar{q}^o \cdot \bar{i}^o$, where $\bar{e}^o$ denotes average emissions of offshoring firms, defined as:

$$\bar{e}^o = \int_{\varphi^o}^{\infty} e^o(\varphi)\frac{g(\varphi)}{1-G(\varphi^o)}d\varphi. \tag{B.50}$$

Solving the integral as in previous steps yields:

$$\bar{e}^o = e^d(\varphi^d)\frac{t}{t^*}\frac{1+\chi^{\frac{\sigma-1}{k}}}{\chi^{\frac{\sigma-1}{k}}}\frac{k}{k-\sigma+1}. \tag{B.51}$$

Substituting $e^d(\varphi^d) = i^d(\varphi^d)q^d(\varphi^d)$ and the respective expressions from cost minimization, we obtain:

$$\bar{e}^o = \frac{k(\sigma-1)}{k-\sigma+1}\alpha(1-\eta)\frac{1+\chi^{\frac{\sigma-1}{k}}}{\chi^{\frac{\sigma-1}{k}}}\frac{w}{t^*}. \tag{B.52}$$

Since $dw/dt < 0$ and the χ-term is decreasing in χ, we conclude that $d\bar{e}^o/dt < 0$.[56] Together with the result $d\bar{i}^o/dt > 0$ and the identity $\bar{e}^o = \bar{q}^o\bar{i}^o$, it follows that average output of offshoring firms declines in response to a unilateral tax increase, i.e., $d\bar{q}^o/dt < 0$.

B.4. Proof of Proposition 3

B.4.1. Effect on aggregate income

Aggregate income in the source country consists of operating profits (joint income of managers and offshoring consultants), worker income, and emissions tax revenues: $I = \bar{\pi}M + wL + tE$. In contrast, host-country income includes only worker income and emissions tax revenues: $I^* = w^*N^* + t^*E^*$.

Using the CES demand structure and Cobb–Douglas production technology in Eqs. (1) and (3), we express income levels as shares of global income Y, which equals total final goods output (with price normalized to one). Under constant markup pricing, operating firm profits account for a fixed share $1/\sigma$, while the remaining share $(\sigma-1)/\sigma$ is split between labor and emissions tax payments according to their factor shares. This yields:

$$I = \left[\frac{1}{\sigma} + \frac{\sigma-1}{\sigma}\left(\gamma^l + \gamma^e\right)\right]Y \qquad \text{and} \qquad I^* = \left[\frac{\sigma-1}{\sigma}\left(\gamma^{l*} + \gamma^{e*}\right)\right]Y. \tag{B.53}$$

Hence, global income is distributed across countries according to revenue shares. While the source country's profit share remains fixed, its labor and tax income shares fall with rising offshoring, whereas those in the host country increase.

As shown above in Appendix B.1.3, global income declines in response to a higher emissions tax in the source country, i.e., $dY/dt < 0$. The impact on the source country's income is further amplified by declining labor and emissions tax shares, $d\gamma^l/d\chi < 0$ and $d\gamma^e/d\chi < 0$. Intuitively, an emissions tax increase makes production more costly, increasing abatement effort

[55] Depending on the initial cross-country tax differential, this increase may occur in an equilibrium where offshoring firms remain cleaner than domestic firms on average, i.e., $\bar{i}^o < \bar{i}^d$. This extends part (i) of Lemma 1 to the aggregate level, making use of the fact that $\bar{\varphi}^o > \bar{\varphi}^d$ due to the self-selection of the most productive firms into offshoring.

[56] In a similar vein, one can show that average emissions of purely domestic firms also decline in t, i.e., $d\bar{e}^d/dt < 0$, although we do not use this result in our analysis.

and reducing output. The most productive firms also shift parts of their production abroad, further reducing the domestic revenue base. As a result:

$$\frac{dI}{dt} = \left[\frac{1}{\sigma} + \frac{\sigma-1}{\sigma}\left(\gamma^l + \gamma^e\right)\right]\underbrace{\frac{dY}{dt}}_{<0} + \left[\frac{1}{\sigma} + \frac{\sigma-1}{\sigma}\left(\underbrace{\frac{d\gamma^l}{dt}}_{<0} + \underbrace{\frac{d\gamma^e}{dt}}_{<0}\right)\right]Y < 0. \tag{B.54}$$

By contrast, the host country benefits from increased offshoring since the production shift leads to higher revenue shares falling on the host country. Mathematically, the increase in income results from rising wage rates w^* and emissions E^*, while the number of workers N^* and the emissions tax t^* remain constant (by assumption). These changes raise the revenue shares γ^{l*} and γ^{e*}, such that the positive shift in the income share dominates the overall decline in global income.

Taken together, the reform generates the following ranking of income changes:

$$\frac{dI}{dt} < \frac{dY}{dt} < 0 < \frac{dI^*}{dt}, \tag{B.55}$$

which confirms the convergence in aggregate income across countries as stated in part (i) of Proposition 3.

B.4.2. Effect on between-country inequality

To evaluate between-country inequality, we compare the minimum post-transfer income across the two countries. In both countries, workers receive wage income and a uniform lump-sum transfer from domestic emissions tax revenues. Accordingly, we define the inequality measure

$$\Xi \equiv \frac{w+b}{w^*+b^*}. \tag{B.56}$$

From Proposition 1, we know that the wage in the source country w declines with the emissions tax, while the host-country wage w^* rises. In addition, changes in emissions tax payments lead to a lower transfer b in the source country and a higher transfer b^* in the host country. Both effects – the falling wage and transfer in the source country, and their corresponding rise in the host country – jointly reduce the inequality measure Ξ, confirming the result in part (i) of Proposition 3.

B.4.3. Effect on inter-group inequality in the source country

To examine inter-group inequality within the source country, we compare the post-transfer income of the average manager and a wage-earner – comprising both workers and offshoring consultants – with each individual receiving the same lump-sum transfer b. The inequality measure is defined as:

$$\Theta \equiv \frac{\bar{\pi} - \chi w + b}{w+b}. \tag{B.57}$$

In the absence of the transfer b, inter-group inequality rises directly in χ. A higher transfer, by contrast, lowers inter-group inequality, as workers benefit relatively more than managers from redistribution. However, the transfer b declines in the emissions tax rate due to the contraction in the domestic tax base, amplifying the increase in Θ. Using the expressions for average profits and the wage rate, Eqs. (A.23) and (A.24), and imposing the government budget constraint $b = tE/N$, we can rewrite Θ as a function of the share of offshoring firms χ, which is the only endogenous variable:

$$\Theta = \frac{[k+\chi(\sigma-1)][k-\sigma+1+\gamma^l k(\sigma-1)] + \gamma^e k(\sigma-1)(k-\sigma+1)}{(k-\sigma+1)[k-\sigma+1+\gamma^e k(\sigma-1)+\gamma^l k(\sigma-1)]}. \tag{B.58}$$

This expression makes clear that inter-group inequality rises with the emissions tax: both the decline in the transfer b and the increasing wedge between manager and worker incomes (net of transfers) contribute to a higher value of Θ. Formally, we obtain $d\Theta/dt > 0$, confirming the result in part (ii) of Proposition 3.[57]

Appendix C. Border carbon adjustment as a complementary policy

C.1. Proof of Proposition 4

C.1.1. Effect on offshoring equilibrium

The implementation of a BCA reduces the cost advantage of offshoring by raising the effective emissions cost for foreign production. This causes a downward shift in the labor market constraint $B(\tau, t, t^*, \chi)$, as shown in Eq. (14). The resulting decline in the equilibrium cutoff κ and the share of offshoring firms χ reflects a direct effect of the policy: fewer firms find it profitable to offshore production. This adjustment is partially offset by an endogenous increase in the relative wage w/w^*, which changes to satisfy the offshoring indifference condition $A(\chi)$, as given in Eq. (12).

Fig. C.1 illustrates the resulting shift in equilibrium. The dotted line shows the downward shift in $B(\cdot)$ caused by the BCA, in contrast to the upward shift (dashed–dotted line) induced by the unilateral emissions tax increase discussed in Section 3.

Moreover, following the argument and proof of Proposition 1, part (i), it follows that the mass of offshoring firms declines. Taken together, this confirms part (i) of Proposition Proposition 4.

[57] This pattern of rising inequality is also illustrated in the source country's Lorenz curve, derived and plotted in Section X.3 of the Online Appendix.

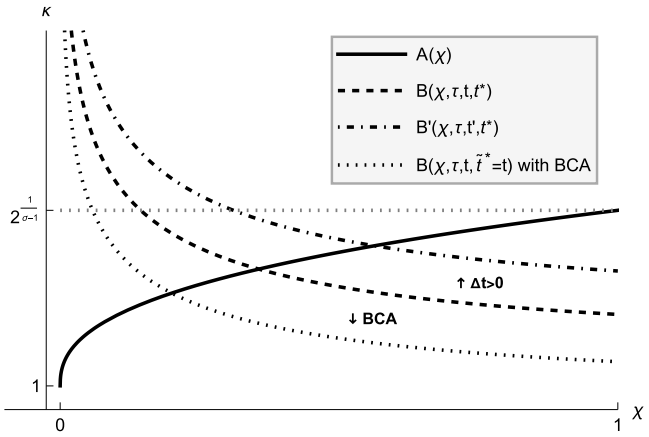


Fig. C.1. The impact of BCA on the offshoring equilibrium.
Notes: The dotted line indicates the shift in $B(\cdot)$ caused by the BCA. The dashed–dotted line reflects the shift induced by the unilateral emissions tax increase discussed in Section 3.

C.1.2. Effect on aggregate emissions

We analyze global emissions using Eqs. (16) and (B.27), where global emissions are defined as $E^W = E + E^*$. Inserting the respective expressions for γ^e and γ^{e*}, and applying $\tilde{t}^* = t$, we obtain:

$$E^W_{BCA} = \alpha(1-\eta)\frac{1}{t}\frac{\sigma-1}{\sigma}Y_{BCA}, \tag{C.1}$$

where the subscript BCA denotes an equilibrium in the presence of a BCA. Forming the ratio of global emissions under the baseline and the BCA regime yields:

$$\frac{E^W}{E^W_{BCA}} = \frac{1+\frac{t}{t^*}\chi+\left(\frac{t}{t^*}-1\right)\chi^{\frac{k-\sigma+1}{k}}}{1+\chi}\frac{Y}{Y_{BCA}} > 1. \tag{C.2}$$

Both terms in this product exceed one, reflecting the direct and indirect effect of a higher effective emissions tax in the host country. The first term captures the shift in revenue shares, while the second reflects the decline in global output with a BCA. This output reduction arises from both a lower share of offshoring – illustrating the indirect effect of the higher effective host country emissions tax – and higher production costs for all remaining offshoring firms. Together, these results confirms that global emissions are lower under the BCA.

From Eq. (B.27), it follows directly that host-country emissions E^* decline. All underlying channels reinforce this result: First, γ^{e*} reduces due to the BCA-induced decline in χ. Second, the relevant emissions tax rises from t^* to $\tilde{t}^* = t$. Third, the decline in global income implies a decrease in the global level of production. By contrast, the impact on emissions in the source country, E, is a priori ambiguous: while global income Y falls, the domestic emissions share γ^e increases. However, combining the previous results shows that E^* declines more strongly than E^W, implying that $E = E^W - E^*$ must increase.

Taken together, this confirms part (ii) of Proposition 4.

C.1.3. Effect on income inequality

We analyze the BCA's impact on income inequality both across and within countries, using the measures introduced in Eqs. (B.56) and (B.57).

The implementation of a BCA alters between-country inequality to:

$$\Xi_{BCA} = \frac{w+\frac{tE+\hat{t}E^*}{N}}{w^*+\frac{t^*E^*}{N^*}}. \tag{C.3}$$

We clearly have $\Xi_{BCA} > \Xi$ for two reasons: First, even in the absence of a BCA transfer from the host to the source country, Ξ rises due to the BCA-induced decline in χ. Second, the source country now collects additional revenues $\hat{t}E^*$, which further increase per-capita transfers. Both mechanisms raise the post-transfer income of the source country relative to the host.

Inter-group inequality, in turn, adjusts to:

$$\Theta_{BCA} = \frac{\bar{\pi}-\chi w+\frac{tE+\hat{t}E^*}{N}}{w+\frac{tE+\hat{t}E^*}{N}}. \tag{C.4}$$

This expression makes clear that $\Theta_{BCA} < \Theta$. First, abstracting from the BCA transfer, the BCA-induced decline in χ translates into a decline in Θ, see Eq. (B.58). Second, the increase in the source country's per-capita transfer due to additional BCA revenues $\hat{t}E^*$ disproportionately benefits wage-earners. This reinforces the decline in inter-group inequality.

Taken together, these results confirm part (iii) of Proposition 4.

C.2. Proof of Proposition 5

C.2.1. Effect on aggregate emissions

In the presence of a BCA, aggregate emissions in the source country and in the host country are given by:

$$E_{BCA} = \frac{\gamma^e}{t}\frac{\sigma-1}{\sigma}Y_{BCA} \qquad \text{and} \qquad E^*_{BCA} = \frac{\gamma^{e*}}{\tilde{t}^*}\frac{\sigma-1}{\sigma}Y_{BCA}. \tag{C.5}$$

Since the environmental policy reform does not affect the share of offshoring, the income shares γ^e and γ^{e*} remain unchanged. Absent this channel, the policy reform $dt = d\tilde{t}^* > 0$ reduces E_{BCA} and E^*_{BCA} directly and indirectly via Y_{BCA}.[58] For the direct negative effect of t on Y, see Eq. (B.14). Hence, with a BCA in place, aggregate emissions decline in both countries, and global emissions unambiguously fall, confirming the result in part (ii) of Proposition 5.

C.2.2. Effect on income and income inequality

We begin by analyzing how the BCA transfer evolves in response to an increase in the source country's emissions tax t, as this is central to determining income and inequality effects.

Total emissions tax payments related to host country emissions are split between a part collected by the source country, the BCA transfer $\hat{t}E^*$, and a part remaining in the host country, t^*E^*. The BCA transfer is given by:

$$\hat{t}E^* = \frac{\hat{t}}{\tilde{t}^*}\gamma^{e*}\frac{\sigma-1}{\sigma}Y. \tag{C.6}$$

Differentiating with respect to t, where $dt = d\tilde{t}^* > 0$, and using Eq. (B.14), yields:

$$\frac{d(\hat{t}E^*)}{dt} = \frac{t^*}{\tilde{t}^{*2}}\gamma^{e*}\frac{\sigma-1}{\sigma}Y - \frac{\hat{t}}{\tilde{t}^*}\gamma^{e*}\frac{\sigma-1}{\sigma}\frac{1}{t}\frac{\alpha(1-\eta)}{1-\alpha(1-\eta)}Y. \tag{C.7}$$

Since $\tilde{t}^* = t$, the BCA transfer increases in t if and only if:

$$\frac{t^*}{\hat{t}} > \frac{\alpha(1-\eta)}{1-\alpha(1-\eta)}. \tag{C.8}$$

Conversely, if this inequality is reversed, the transfer decreases in response to the tax increase. Thus, the effect of an emissions tax increase on the BCA transfer is generally ambiguous. It depends on the initial tax differential between the two countries, and the cost shares of emissions and non-routine labor in production. Specifically, the transfer increases in t when the initial tax differential is sufficiently small – i.e., when $t^* > \hat{t}$.

The BCA transfer divides the host country's emissions tax revenues $\gamma^{e*}(\sigma-1)/\sigma Y$. Specifically, the share $\hat{t}/\tilde{t}^*$ accrues to the source country, while the remaining share $1-\hat{t}/\tilde{t}^* = t^*/\tilde{t}$ remains with the host country. Accordingly, adjusting Eq. (B.53), aggregate income levels of the two countries are given by:

$$I^* = \frac{\sigma-1}{\sigma}\left[\gamma^{l*} + \left(1-\frac{\hat{t}}{\tilde{t}^*}\right)\gamma^{e*}\right]Y \qquad \text{and} \qquad I = \left[\frac{1}{\sigma} + \frac{\sigma-1}{\sigma}\left(\gamma^l + \gamma^e + \frac{\hat{t}}{\tilde{t}^*}\gamma^{e*}\right)\right]Y, \tag{C.9}$$

which implies an income ratio of:

$$\frac{I}{I^*} = \frac{\left[\frac{1}{\sigma} + \frac{\sigma-1}{\sigma}\left(\gamma^l + \gamma^e + \frac{\hat{t}}{\tilde{t}^*}\gamma^{e*}\right)\right]}{\frac{\sigma-1}{\sigma}\left[\gamma^{l*} + \left(1-\frac{\hat{t}}{\tilde{t}^*}\right)\gamma^{e*}\right]}. \tag{C.10}$$

From these expressions, we derive three key implications of a unilateral environmental policy reform under a BCA on income: First, host country income, I^*, declines. This follows from the decline in aggregate output Y, while the revenue shares γ^{l*} and γ^{e*} remain constant. Furthermore, the share of aggregate emissions tax income remaining in the host country falls, as the respective ratio $t^*/\tilde{t}^*$ declines. Second, the income ratio, I/I^*, increases. Since the income γ^l, γ^e, γ^{l*}, and γ^{e*} are constant, and the source country's share of foreign emissions revenues $\hat{t}/\tilde{t}^*$ increases, the source country's income rises relative to that of the host country. Third, the effect on source country income, I, is ambiguous. While global output declines, the source country gains from the BCA transfer. The net effect depends on whether this transfer increase is strong enough to offset the decline in Y – see Eq. (C.8) for the condition of a rise in the BCA transfer to begin with.

The between-country inequality measure, as defined in Eq. (B.56), becomes:

$$\Xi = \frac{w + tE + \hat{t}E^*}{w^* + t^*E^*}. \tag{C.11}$$

An increase in the BCA transfer raises the numerator by increasing post-transfer income in the source country, while reducing the denominator by shrinking the share of emissions revenues retained in the host country. Both effects push Ξ upward, implying that between-country inequality rises when the BCA transfer increases. Conversely, if the BCA transfer declines, the inequality response

[58] If $\gamma^e < \gamma^{e*}$, the reform reduces host country aggregate emissions more strongly than source country emissions in absolute terms. This requires χ to be sufficiently large.

is ambiguous. On the one hand, a falling transfer reduces source-country income and increases host-country income – the mirror image of the rising-transfer case – which tends to lower Ξ. On the other hand, both countries experience a decline in their income, abstracting from the transfer, due to the emissions tax. This increases the relative weight of the redistributed BCA transfer, putting upward pressure on Ξ.

The measure of source country's inter-group inequality, as defined in Eq. (B.57), becomes:

$$\Theta = \frac{\bar{\pi} - \chi w + tE + \hat{t}E^*}{w + tE + \hat{t}E^*}. \tag{C.12}$$

Adding the same amount to both numerator and denominator decreases the ratio when the denominator is smaller (as is the case here, since worker income is low relative to total managerial surplus). Therefore, Θ decreases if the BCA transfer increases. If the transfer falls, the outcome again becomes ambiguous. On the one hand, a falling transfer increases Θ directly by disproportionately affecting the lower-income group – the mirror image of a rise in the transfer. On the other hand, both managerial and worker income net of transfers decline with the emissions tax, which raises the relative weight of the transfer in total income. This induces a mitigating force on Θ, even if the transfer itself does not increase.

Taken together, these results confirm part (iii) of Proposition 5.

Appendix D. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jinteco.2025.104185.

Data availability

Replication Package: Unilateral Environmental Policy and Offshoring (Original data) (Mendeley Data)

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